RIVERSCAPE GENETICS IDENTIFIES A CRYPTIC LINEAGE OF SPECKLED DACE *(RHINICHTHYS OSCULUS)* IN THE KLAMATH-TRINITY BASIN

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A Thesis Presented to

The Faculty of Humboldt State University

In Partial Fulfillment of the Requirements for the Degree

Master of Science in Natural Resources: Fisheries

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December 2014

ABSTRACT

RIVERSCAPE GENETICS IDENTIFIES A CRYPTIC LINEAGE OF SPECKLED DACE *(RHINICHTHYS OSCULUS)* IN THE KLAMATH-TRINITY BASIN

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Cataloging biodiversity is of utmost importance given that habitat destruction has dramatically increased extinction rates. While the presence of cryptic species poses challenges for biodiversity assessment, molecular analysis has proven useful in uncovering this hidden diversity. Using nuclear microsatellite markers and mitochondrial DNA (mtDNA), I investigated the genetic structure of Klamath speckled dace (*Rhinichthys osculus klamathensis*), a subspecies endemic to the Klamath-Trinity Basin. Analysis of 25 populations within the basin uncovered a cryptic lineage of speckled dace restricted to the Trinity River system and a possible contact zone between this lineage and speckled dace occurring in the Klamath Basin. The extent of mtDNA divergence between the Klamath and Trinity speckled dace was consistent with levels observed between sibling fish species (~3%). Trinity River system populations exhibited significantly lower levels of microsatellite genetic diversity (H_e=0.49, Ar=9.46) compared to Klamath Basin populations (H_e=0.64, Ar=12.35). Levels of hybridization between the lineages appeared to be low and confined to areas near the confluence of the Klamath and Trinity Rivers. The deep divergence between the lineages suggests that historical biogeographical processes are responsible. The precise mechanism that generated these lineages and currently maintains them as distinct in the absence of

physical barriers is unknown. This study highlights the importance of incorporating molecular analysis into biodiversity research to uncover cryptic diversity.

ACKNOWLEDGEMENTS

First and foremost I would like to thank my graduate advisor, Dr. Andrew Kinziger, who fully supported me and always had time for my many questions while working on this project. Thank you for believing in me, for challenging me, and for the long nights you spent editing my drafts. Without your wisdom, mastery of genetics and enthusiasm, this thesis would have been impossible.

I am indebted to Damon Goodman for his assistance in the field and help facilitating this research. Damon's extensive knowledge of the Klamath-Trinity Basin was invaluable to this project. Thank you to Rod Nakamoto and Dr. Bret Harvey for laying the groundwork that this project is based on, and to Bill Tinniswood for his help in Oregon. I would also like to thank my committee members, Dr. Tim Mulligan and Dr. Bret Harvey, for their expert advice and review of my draft.

Thank you to the amazing Dana Herman for her technical assistance and emotional support, without her encouragement I couldn't have done it. Thanks to Dr. Deb Duffield for giving me a chance and starting me on my path. Thank you to my speckled dace collectors: Conrad Newell, Sam Rizza, Robbie Mueller and to my fantastic lab mate Molly Schmelzle. To Jeff Abrams for showing me the ropes when I first got to HSU. Thanks to Tom Huteson, for his help with images and to Chloe Joesten for help with DNA extractions. To Marin Rod and Gun Club and Geoffrey Bain Memorial scholarship for their generous financial support.

A special thanks to my mother and father for their love and support throughout my life. Thank you for raising me to be curious and teaching me to value the natural world.

DEDICATION

This thesis is dedicated to the memory of my mother, Patricia Hursh.

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INTRODUCTION

Biodiversity is being lost at an astonishing rate due to forces associated with human population growth, habitat alteration and global climate change (McNeely et al 1990; Chapin III et al. 2000). The situation is particularly troublesome for freshwater ecosystems, as freshwater biodiversity has declined at a faster rate compared to terrestrial and marine systems (Ricciardi and Rasmussen 1999; Jenkins 2003). This disproportionate decline in freshwater biodiversity may be due to the large proportion of earth's biodiversity that freshwater ecosystems harbor. Freshwater ecosystems cover only 0.8% of the earth's surface and contain a mere 0.01% of the world's water by volume, yet, these systems are inhabited by 6% of all described species (Dudgeon et al. 2006). Freshwater ecosystems are also heavily impacted by anthropogenic effects as humans live disproportionally near waterways.

Fishes are the most-studied indicators of biodiversity decline in freshwater ecosystems and possess significant cultural and economic value (Moyle et al. 2011). Despite this, our knowledge and understanding of freshwater fish diversity, patterns of endemism, and genetic variation is limited, with many fish species yet to be described (Myers et al. 2000; Abell et al. 2008). As species continue to disappear, accompanied by the uncertainty of global climate change, it is imperative that we document biodiversity in order to prioritize conservation efforts, predict extinction rates and set standards for monitoring ecosystem change (Pimm et al. 1995).

The International Union for Conservation of Nature (IUCN) recognizes three primary levels of global biodiversity: genes, species and ecosystems. Which level of biodiversity receives priority for conservation has been a contentious debate across numerous biological disciplines (Bowen 1999). Species have been considered to be the foundation of biodiversity (McNeely et al. 1990), and the standard metric to monitor environmental conditions (Noss 1990). Bias towards the use of species to monitor and inventory biodiversity has historically been out of necessity, as species are relatively easy to study, compared to genes or ecosystems. However, with the advent of high-throughput DNA sequencing and the realization that a loss of genetic diversity comes with very serious risks for populations (Frankel 1974; Frankham 2005), the importance of genes has been increasingly recognized in conservation (Avise 1998; DeSalle and Amato 2004).

For freshwater fish conservation, molecular analysis has provided numerous ways to study population structure, identify threats to biodiversity and solve taxonomic difficulties (Vrijenhoek et al. 1998 and citations therein). DNA-based research has also facilitated conservation at the highest level, the ecosystem, by identifying areas of endemism and by prioritizing targets for marine and freshwater reserve planning (DeSalle and Amato 2004; Cook et al. 2008). However, given the important role that genetic variation plays in the support of species and ecosystem conservation, direct action and international policy to preserve diversity at the genetic level is still greatly lacking (Laikre 2010).

Defining species based solely on visual and ecological differences carries inherent risk, as speciation is not always followed by morphological change and convergent

evolution can produce similar forms (Avise 2000). The use of molecular techniques to examine genetic diversity and identify new species is becoming increasingly common, as technological advances have made DNA sequencing both efficient and affordable (Hajibabaei et al. 2007). With the increase in DNA-based studies, cryptic species are rapidly being discovered in many different taxa and bioregions (Pfenninger and Schwenk 2007). A cryptic species can be defined as two or more morphologically similar species mistakenly classified as one species, though they are often reproductively isolated (Yoder et al. 2002) and can have a deep genetic divergence (Ward et al. 2005).

Cryptic species pose a substantial challenge to taxonomy and policy. However, their discovery and classification provides a more accurate representation of the planet's biodiversity. Furthermore, the identification of cryptic species is important for correctly recognizing species invasions (Bucciarelli et al. 2002), making conservation plans (Stuart et al. 2006), defining species (Herbert et al. 2004) and the study of evolutionary processes (Mendelson and Shaw 2002). While many cryptic species are living in sympatry and do not exchange genes (Bickford et al. 2007), other cryptic species can (Tringo et al. 2013), providing a window into the mechanisms responsible for maintaining species boundaries (Harrison 1993). For example, a cryptic hatchetfish *(Carnegiella meathae)* in the Amazonian Floodplain was found to be three different species; while two of the species could produce hybrids, the third, rarer species was found to be reproductively isolated (Piggot et al. 2011).

One species with the potential to harbor a number of cryptic species is the speckled dace (*Rhinichthys osculus*). The speckled dace is a small (80 mm standard

length) freshwater cyprinid that is one of most widespread native fish in the western United States (Lee et al. 1980). Speckled dace have a highly variable morphology throughout their range that is reflected by the wide assortment of ecosystems that they inhabit (Moyle 2002). The variety of distinct forms of speckled dace and wide distribution of the species has led to a confusing and complicated taxonomic history. Presently, speckled dace are considered to be a single, wide ranging, variable species containing a number recognized and unrecognized subspecies some of which are listed (Moyle 2002). Recently, molecular assessments have shown that speckled dace represent a polyphyletic taxon containing a number undescribed taxa (Ardren et al. 2010; Hoekzema and Sidlauskas 2014).

A number of phylogenetic studies have undertaken the task of untangling the evolutionary history of speckled dace (Oakey et al. 2004; Pfrender et al. 2004; Smith and Dowling 2008; Ardren et al. 2010; Billman et al. 2010; Hoekzema and Sidlauskas 2014). These studies have shown that deep genetic divergence generally occurs among speckled dace isolated in separate river basins. Diversification of speckled dace, like many North American freshwater fishes, can be attributed to complex geologic and climatic processes that caused long periods of isolation between basins, interspersed with episodes of dispersal through drainage rearrangements (Minckley et al. 1986).

The taxonomy and systematics of speckled dace are complicated by the potential for hybridization. Hybridization is common in freshwater fishes, especially between cyprinids, due to external fertilization, limited availability of spawning sites, and weak behavioral isolating mechanisms (reviewed in Scribner et al. 2001). However, in some

instances strong selection against hybrid offspring has been reported among cyprinids (Dowling and Moore 1985). Genetic studies investigating hybridization in speckled dace and other species are limited, but evidence of hybridization between introduced speckled dace and relict dace (*Relictus solitarius*) in the Great Basin has been reported (Houston et al 2012). Also, Smith (1973) suggested hybridization between speckled dace and two other species (longnose dace *Rhinichthys cataractae* and redside shiners *Richardsonius balteatus*) based upon morphological analyses.

This research investigated speckled dace throughout the Klamath-Trinity Basin in south central Oregon and northwestern California. The Trinity River system, which includes the Trinity River and its tributaries, is hereafter referred to as the "Trinity" and the Klamath River system, minus the Trinity River, is hereafter referred to as the "Klamath". Speckled dace occurring in the Klamath-Trinity Basin are recognized as an endemic subspecies Klamath speckled dace (*R. o. klamathensis*) are nearly continuously distributed throughout the entire Klamath-Trinity Basin and occur at high abundance in many areas (Moyle 2002). The presence of speckled dace in the Lower Klamath and Trinity Rivers is somewhat unexpected, as the geology of the area has made colonization by freshwater dispersers difficult and the species is absent from most coastal northern California streams (Moyle 2002)

The Klamath subspecies designation is supported by deep genetic divergence of Klamath speckled dace from outside basins (Oakey et al. 2004; Pfrender et al. 2004; Kinziger et al. 2011). However, two studies have resolved evidence for multiple genetically distinct speckled dace lineages within the Klamath-Trinity Basin. Kinziger et

al. (2011) found deep divergence between speckled dace occurring in the Klamath versus those from the Trinity and Pfrender et al. (2004) uncovered evidence for two sympatric lineages of speckled dace in the Upper Klamath. However, the number of localities and genetic markers examined to date is insufficient to provide a clear understanding of speckled dace lineages within the Klamath-Trinity Basin.

The objective of this study was to conduct a comprehensive assessment of speckled dace genetic structure in the Klamath-Trinity Basin using multiple genetic markers (nuclear microsatellites and mtDNA) to identify cryptic lineages, their geographic boundaries, contact zones and levels of hybridization. I discuss the historical processes that may have played a role in the development of the genetic structuring patterns recovered within Klamath-Trinity River speckled dace and the conservation and taxonomic implications of the results.

MATERIALS AND METHODS

Field Collections

The Klamath-Trinity Basin (41,000 km^2) has two distinct regions which are quite different in terms of climate and habitat, the upper basin and lower basin, divided by Iron Gate Dam. In the rain shadow of the Cascade Range, the upper Klamath Basin is characterized by flat valleys, slow rivers, and large lakes. In contrast, the lower basin is characterized by higher levels of precipitation, more mountainous terrain, and swift, frigid rivers that are predominately confined to bedrock canyons. The Trinity River system in the lower basin is the largest tributary to the Klamath River (7680 km^2), comprising 19% of the entire basin and containing some of the highest elevations and steepest gradients within the entire basin. Speckled dace were collected at 25 locations throughout the Klamath-Trinity Basin (Fig. 1). Klamath River system sites (n=16) spanned from the mouth of the Klamath River to above Klamath Lake. Trinity River system sites (n=9) included locations near the confluence of the Trinity and Klamath Rivers to above Trinity Lake (Table 1). Collection locations were selected to provide broad geographic coverage, include sites which fell above and below barriers (e.g., dams and large waterfalls) and to include all major tributary systems. Fine-scale sampling at a broad geographic range is crucial for discerning genetic structure in organisms with restricted dispersal ability, coupled with localized genetic differentiation (Young et al. 2013). Specimens were collected using seine nets, or a backpack electrofisher and were

supplemented by specimens from Kinziger et al. (2011) which were archived at the Humboldt State University (HSU) Fish Collection. Specimens were euthanized, using an overdose of tricaine methanesulfonate (MS-222) and vouchered whole, or were anesthetized with MS-222 and then a small caudal fin clip was collected before releasing the fish. Whole specimens/tissue were preserved in 95% ethanol and deposited into the HSU Fish Collection until DNA extraction. Speckled dace were collected for this study under a scientific collecting permit in California (SC-12220) and Oregon (17382), following an Institutional Animal Care and Use permit (11/12.F30.A).

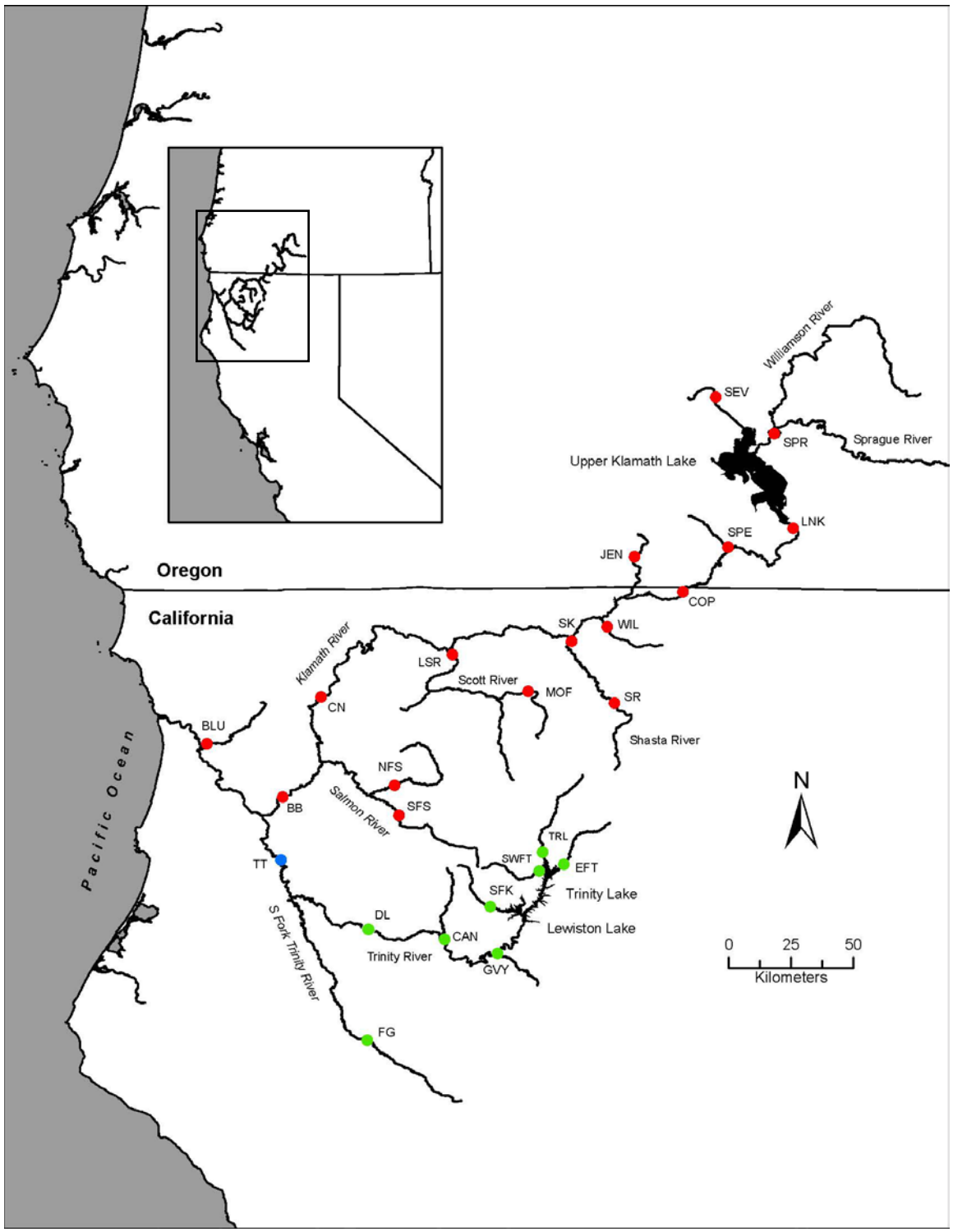

Figure 1. Graphical depiction of the 25 collection locations for Klamath-Trinity Basin speckled dace, including the Klamath River system (red), Trinity River System (green), and Tish Tang Creek (TT) (blue). Site abbreviations as in Table 1.

Table 1. River system, population, site abbreviation (ID), latitude, longitude, collection date, and the Humboldt State University Fish Collection numbers (HSU ID) from Klamath-Trinity basin speckled dace.

River System	**Population**	**ID**	**Latitude**	**Longitude**	**Date**	**HSU ID**
Klamath						
	Seven Mile Creek	SEV	42.69421	-122.0727	9/2/2012	5079/5080
	Sprague River	SPR	42.56339	-121.8624	7/25/2012	5068
	Link River	LNK	42.2216	-121.7934	9/2/2012	5077/5078
	Spencer Creek	SPE	42.15257	-122.0278	6/23/2012	5066
	Jenny Creek	JEN	42.11791	-122.367	6/23/2012	5065
	Klamath River (above Copco 2)	COP	41.9915	-122.1902	7/26/2012	5002
	Willow Creek	WIL	41.86591	-122.4648	7/25/2012	5001
	Shasta River (Upper)	SR*	41.591	-122.438	2/5/2010	4943
	Shasta River (Lower)	SK	41.81309	-122.5923	7/26/2012	4999
	Moffet Creek	MOF	41.6337	-122.749	7/26/2012	5010
	Lower Scott River	LSR	41.76529	-123.0213	7/1/2012	5131
	Klamath River (Coon Creek)	CN	41.61363	-123.4958	8/31/2012	5076
	Klamath River (Big Bar)	BB	41.25204	-123.6348	8/31/2012	5073
	Blue Creek	BLU	41.44417	-123.9069	5/31/2012	4998
	North fork Salmon	NFS	41.29301	-123.2301	7/25/2012	5036
	South Fork Salmon	SFS	41.18689	-123.2139	7/25/2012	5014
Trinity						
	Tish Tang Creek	TT	41.02586	-123.6403	7/24/2012	5023/5083
	Trinity River (Del Loma)	DL	40.7749	-123.325	7/31/2013	5123
	South Fork Trinity River	FG*	40.377	-123.3256	10/25/2004	4940
	Canyon Creek	CAN*	40.738	-123.0495	8/10/2010	4945/4944
	Grass Valley Creek	GVY	40.68979	-122.8576	7/24/2012	5007
	Stuart Fork Creek	SFK	40.8558	-122.8851	7/24/2012	5030
	Swift Creek	SWFT	40.98642	-122.7089	7/25/2012	5012
	Trinity River (above Trinity Lake)	TRL	41.05328	-122.6965	7/25/2012	5004
	East Fork Trinity	EFT	41.0085	-122.6201	7/26/2012	5016

*Samples analyzed in Kinziger et al. (2011).

Microsatellite Genotyping Methods

Speckled dace were genotyped at nine microsatellite loci, however, preliminary tests indicated that the locus CypG 13 consistently departed from Hardy-Wienberg expectations and therefore was removed. The final data set contained eight microsatellite loci (Table 1). The microsatellite loci used in this study were based on previous speckled dace genetic studies in California (Kinziger et al. 2011) and Oregon (Hoekzema and Sidlauskas 2014). Whole genomic DNA was extracted from fin tissue using the Chelex DNA extraction method under the manufacturer's recommendations and amplified via the polymerase chain reaction. Amplifications were performed as either 10 or 12.7-µl reactions using GoTaq Colorless Master Mix (Promega, Madison, WI) in a MJ Research (Waltham, MA) PTC-100 or an Applied Biosystems (Grand Island, NY) 2720 thermal cycler. The forward primer of each primer pair was labeled with a WellRED fluorescent dye (Sigma-Aldrich, St Louis, MO) for identification. Products were visualized and allele sizes determined with the Beckman-Coulter CEQ 8000 Genetic Analysis System (Brea, CA). Allele sizes were scored twice and any discrepancies were either resolved or the genotype was removed. Individuals missing more than two loci from their multi-locus genotype were removed from the dataset. Tests for conformance to Hardy-Weinberg equilibrium (HWE) and linkage disequilibrium were conducted using GENEPOP V4.2 (Raymond and Rousset 1995). Loci were checked for null alleles, stutter peaks, and large allele drop-out using MICRO-CHECKER v 2.2.3 (Van Oosterhout et al. 2004).

Genetic Analysis

Nuclear diversity

Observed heterozygosity (H_o), Hardy-Weinberg expected heterozygosity (H_e), and allelic richness (A) were calculated in ARLQUIN 3.11 (Schneider et al. 2000). Rarified allelic richness (A_R) and private allelic richness (A_p), both standardized to a sample size of 48 genes, were calculated using HP-Rare (Kalinowski 2005). Permutation tests (2,000 replicates) for significant differences in A_R and H_e between groups were conducted using the software FSTAT v2.9.3 (GOUDET 1995).

Population structure

Pairwise estimates of genetic differentiation (F_{ST}) between populations and tests of their significance were conducted using FSTAT v2.9.3 (Goudet 1995). Pairwise estimates of standardized F_{ST}, which is standardized to the largest possible value obtainable, was calculated in GENODIVE (Merimans and Van Tienderen 2004). A graphical depiction of genetic divergence between populations was generated by constructing a neighbor-joining tree of Cavalli-Svorza and Edwards chord distances using the software PHYLIP v3.68 (Felsenstein 2004). Branch support was evaluated by a bootstrap analysis with 1,000 replicates.

Tests for conformance to an isolation-by-distance (IBD) gene flow model were conducted by evaluating the relationship between river distances and genetic distances by conducting a Mantel test (10,000 randomizations) as implemented in the software IBDWS v.3.23 (Jensen et al. 2005). Pairwise river distances (KM) between populations were

calculated in GIS ArcMap 10.1 and genetic distances consisted of standardized pairwise $F_{ST.}$

Two different genetic clustering approaches were employed to analyze the microsatellite dataset as results may sometimes vary according to approach (Latch et al. 2005). The first method, Discriminant Analysis of Principal Components (DAPC) (Jombart et al. 2010), uses a multivariate approach to visualize population differentiation with no assumptions of population genetic models. In DAPC, data are first transformed into uncorrelated variables via Principal Component Analysis (PCA), and then uncorrelated variables from the PCA are analyzed using Discriminant Analysis (DA). The analysis attempts to maximize between group variation among predefined groups. The DAPC analysis was conducted using the R package ADEGENET (Jombart 2008).

The second genetic clustering method was a Bayesian algorithm implemented in the program STRUCTURE v 2.3.4 (Pritchard et al., 2000), which estimates the number of discrete genetic clusters (K) of individuals by minimizing Hardy-Weinberg and linkage disequilibrium. An individual's assignment to each cluster is also calculated, called the admixture proportion (q), and can be used to estimate genetic admixture and introgression. After discarding the first 100,000 steps of the MCMC simulations as burn-in, 100,000 additional steps were performed. A total of 20 iterations were conducted assuming K=1 … 12. To estimate the number of clusters in the data, the *ad hoc* method of ΔK (Evanno et al. 2005) was calculated using STRUCTURE HARVESTER (Earl & vonHoldt 2012), where the largest change in K infers the number of clusters. The Greedy algorithm (10,000 replications) was utilized to merge independent runs at a given K using

the program CLUMP v 1.1.2 (Jackobsen and Rosenberg, 2007). The results from CLUMP were visualized using the software DISTRUCT V 1.1 (Rosenberg 2003).

Hybridization

Results indicated the presence of genetically distinctive groups in the Trinity and Klamath. To estimate levels of hybridization between these groups, *q* values from STRUCTURE (K=2) were used to assign individuals to pure Trinity, pure Klamath or as a hybrid. A threshold of $q > 0.9$ was set for assignment to pure Trinity, $q < 0.1$ for pure Klamath, and $0.1 < q < 0.9$ for hybrids (following Vähä & Primmer, 2006). Only those individuals with 90% probability intervals entirely contained within the designated threshold ranges, 0.9-1 for Trinity, 0-0.1 for Klamath, and 0.1-0.9 for hybrids, were considered categorized as hybrid or pure with high probability. Hybrids whose 90% probability interval was not entirely contained within the threshold ranges (e.g., 0.1 to 0.9) were considered to have equivocal assignments.

Mitochondrial DNA

A subset of 236 speckled dace from all 25 populations (average of 9 individuals per population, range 2-23 individuals) were sequenced for a 548-bp fragment of the mitochondrial cytochrome *b* gene (cyt *b*). DNA amplification was conducted with primers LA and HA (Dowling and Naylor 1997) using the following thermal cycling routine: 35 cycles of 94°C for 60 s, 48°C for 60 s, and 72°C for 120 s. LA and three primers designed for this study were used for sequencing (Table 3). PCR products were purified and sequenced at High-Throughput Sequencing (University of Washington,

Department of Genome Sciences) using an Applied Biosystem 3730 xl sequencer. Chromatograms were visually inspected and manual corrections were made to the sequences. Sequences were aligned in MEGA6 (Tamura *et al.* 2013) using MUSCLE (Edgar 2004).

Mitochondrial DNA diversity and population structure

DNASP v5.10 (Rozas et al. 2003; Librado & Rozas 2009) was used to estimate the number of unique haplotypes (H), the number of variable sites (S), nucleotide diversity (π), haplotype diversity (H_d) and percent sequence divergence (D_{xy}). Uncorrected *p*-distances were calculated as the average number of nucleotide differences between population pairs using MEGA6 (Tamura et al. 2013).

To visualize relationships among the mtDNA haplotypes, a 95% maximum parsimony haplotype network was constructed with TCS 1.21 (Clement et al. 2000). In addition, a maximum-likelihood (ML) tree was generated in MEGA6. Branch support for the tree was estimated via 1000 bootstrap pseudoreplicates. Publicly available speckled dace sequences were added to the sequences from this study to evaluate monophyly of Klamath-Trinity River speckled dace and make comparisons to populations from other basins. The unique haplotypes from the Klamath and Trinity were trimmed to 530 bp to match 15 speckled dace haplotypes that were added (Table 3 in the appendix). *Rhinichthys atratulus* served as an out-group to root the tree (Dowling et al. 2002). The additional sequences were obtained from Genbank and Dryad Digital Repository (doi:10.5061/dryad.ht554, Appendix A). The best model of sequence evolution was Kimura two-parameter model (K80) identified using jModelTest v2.1.4 (Posada 2008)

based upon Akaike's Information Criterion corrected for small sample size AIC_c (Burnham and Anderson 2002).

Table 2. Locus, primers (Forward (F) and Reverse (R)), number of alleles, cycling conditions, size (size range (bp)), and references for the eight microsatellite loci assayed in Klamath-Trinity Basin speckled dace.

Locus	Primer	Alleles	Cycling Conditions	Size	References
CypG3	F:AGTAGGTTTCCCAGCATCATTGT R:GACTGGACGCCTCTACTTTCATA	63	95°C for 60 seconds, 30 cycles of: 95°C for 60 seconds, 67°C for 45 seconds with -0.5°C/cycle, and 72°C for 120 seconds.	187-438	Baerwald and May 2004
CypG9	F:GCAGTCACGTATTAAGGCGAGCAG R: GAGCGGACTCTCAGGCACCTACC	5	95°C for 60 seconds, 30 cycles of: 95°C for 60 seconds, 67°C for 45 seconds with -0.5°C/cycle, and 72°C for 120 seconds.	104-120	Baerwald and May 2004
CypG24	F: CTGCCGCATCAGAGATAAACACTT R: TGGCGGTAAGGGTAGACCAC	25	95°C for 60 seconds, 30 cycles of: 95°C for 60 seconds, 67°C for 45 seconds with -0.5°C/cycle, and 72°C for 120 seconds.	143-194	Baerwald and May 2004
CypG27	F: AAGGTATTCTCCAGCATTTAT R: GAGCCACCTGGAGACATTACT	36	95°C for 3 minutes, 35 cycles of: 95°C for 30 seconds, 57°C for 45 seconds, 72°C for 60 seconds. ext at 72°C for 5 minutes.	215-378	Baerwald and May 2004

Locus	Primer	Alleles	Cycling Conditions	Size	References
CypG33	F: TATGAGCTTTGGAAAGAGACACTG R: AATAGCCGGGAAATTATCAATAGA	5	95°C for 60 seconds, 30 cycles of: 95°C for 60 seconds, 67°C for 45 seconds with -0.5°C/cycle, and 72°C for 120 seconds.	83-96	Baerwald and May 2004
Lco1	F: CACGGGACAATTTGGATGTTTTAT R:AGGGGGCAGCATACAAGAGACAAC	78	94°C for 120 seconds, 30 cycles of: 94°C for 30 seconds, 60C for 30 seconds, 72°C for 30 seconds.	288-562	Turner et al. 2004
Lco4	F: ATCAGGTCAGGGGTGTCACG R: TGTTTATTTGGGGTCTGTGT	17	94°C for 120 seconds, 30 cycles of: 94°C for 30 seconds, 60C for 30 seconds, 72°C for 30 seconds.	229-268	Turner et al. 2004
RHCA 20	F: CTACATCTGCAAGAAAGGC R: CAGTGAGGTATAAAGCAAGG	29	92°C for 30 seconds, 45 cycles of: 92°C for 30 seconds, 50°C for 15 seconds, 68°C for 5 seconds. ext at 68°C for 120 seconds.	87-156	Girard and Angers 2006

Table 3. Primers used for mtDNA cyt b sequencing of Klamath-Trinity Basin speckled dace. LA and HA (Dowling and Naylor 1997) were used for PCR amplification and LA, KB1, KB2, and HD-ALT were used to sequence the 548 bp fragment.

Primer	Sequence
LA	5' - GTGACTTGAAAAACCACCGTTG - 3'
HA	5' - CAACGATCTCCGGTTTACAAGAC - 3'
KB1	'5-GTCAGGGATGTGAGGGCTAA-3'
KB2	'5-GGGGTAAAATTTTCTGGATCG-3
HD-ALT	'5-GGGTTGTTTGATCCCGTTTCGT-3'

RESULTS

Microsatellites

A total of 1075 individuals were assayed in the microsatellite dataset with an average of 43 fish per location (range 30-48). The microsatellite loci were highly polymorphic, containing a total of 258 alleles across 8 loci (mean 32.3 alleles/locus; range 5-78). Out of a total of 200 tests (8 loci and 25 populations) for conformance to Hardy-Weinberg expectations (HWE), nine were significant after Bonferroni correction (Rice 1989) (critical value =0.00025). Departures from HWE were likely due to null alleles, as suggested by MICRO-CHECKER, but heterozygous deficit may also have resulted from the Wahlund effect, or combined collection of genetically differentiated populations. There was no evidence of stuttering, large allele drop-out, or linkage disequilibrium in the loci.

Genetic diversity and population structure

Speckled dace genetic diversity differed between the Klamath and the Trinity. Mean expected heterozygosity (H_e) was 0.70 (range 0.45 to 0.74) in the Klamath and 0.54 in the Trinity (range 0.49 to 0.67, Table 4.). Mean rarified allelic richness (A_R) was 12.35 (7.58 to 14.26) in the Klamath and 9.46 (8.03 to 12.55) in the Trinity. Expected heterozygosity and rarified allelic richness were significantly lower in the Trinity in comparison to the Klamath (P=0.001). The Upper Klamath population, Jenny Creek (JEN), had reduced values of genetic diversity (Ar=7.58, H_e=0.45) compared to nearby

Klamath populations, as did Salmon River populations (NFS: A_R=10.56, H_e=0.62; SFS: A_R=9.19, H_e=0.59). In contrast, Tish Tang Creek (TT) had higher than average allelic richness and heterozygosity (A_R=12.55, H_e=0.67) compared to other Trinity populations.

A total of 257 of 300 pairwise F_{ST} values (0.00-0.345) were significant (P<= 0.000167, Appendix B). Non-significant comparisons tended to occur among populations from within the same basin. Mean standardized F_{ST} for comparisons between Klamath and Trinity was 0.480, whereas the level of divergence between populations within the Klamath (mean 0.052) and Trinity (mean Trinity=0.070) were much lower. A general exception to these patterns was Jenny Creek (JEN), which was resolved as divergent from both the Trinity and Klamath (Table 5).

Tests for isolation-by-distance (IBD) were conducted separately for Klamath and Trinity due to the large divergence between these groups. The relationship between pairwise genetic distances (F_{ST}) and river distance was significant for both the Klamath and Trinity populations ($p < 0.05$) and the intercept was essentially zero in both cases, which is consistent with IBD model of gene flow (Fig. 2).

Table 4. River system, population abbreviation (ID), sample size (n), observed heterozygosity (H_o), expected heterozygosity (H_e), allelic richness (A), rarified allelic richness (A_R), and rarified number of private alleles (A_p).

River System	**Population**	**ID**	**n**	H_o	H_e	A	A_R	A_p
Klamath								
	Seven Mile Creek	SEV	44	0.65	0.74	15.00	12.52	0.15
	Sprague River	SPR	46	0.67	0.74	17.90	14.26	0.32
	Link River	LNK	46	0.70	0.74	14.10	11.82	0.13
	Spencer Creek	SPE	43	0.65	0.74	15.40	12.81	0.05
	Jenny Creek	JEN	48	0.41	0.45	9.40	7.58	0.40
	Copco 2	COP	47	0.66	0.73	16.50	13.24	0.17
	Willow Creek	WIL	44	0.72	0.72	15.40	12.42	0.16
	Shasta River (Upper)	SR	30	0.67	0.69	13.50	12.29	0.22
	Shasta River (Lower)	SK	48	0.72	0.74	17.60	13.64	0.23
	Moffet Creek	MOF	47	0.70	0.72	17.00	13.13	0.09
	Lower Scott River	LSR	47	0.67	0.74	17.40	13.83	0.05
	Coon Creek	CN	44	0.66	0.74	17.80	14.10	0.05
	Big Bar	BB	46	0.63	0.74	15.60	12.76	0.28
	Blue Creek	BLU	47	0.64	0.74	16.90	13.44	0.13
	North fork Salmon	NFS	34	0.54	0.62	11.90	10.56	0.15
	South Fork Salmon	SFS	44	0.53	0.59	11.60	9.19	0.05
	Mean		**44**	**0.64**	**0.70**	**15.19**	**12.35**	**0.16**
Trinity								
	Tish Tang Creek	TT	40	0.60	0.67	14.50	12.55	0.04
	Del Loma	DL	43	0.47	0.52	11.90	9.76	0.01
	South Fork Trinity River	FG	48	0.51	0.55	10.30	8.69	0.12
	Canyon Creek	CAN	31	0.47	0.49	8.50	8.03	0.05
	Grass Valley Creek	GVY	42	0.47	0.54	10.50	9.07	0.04
	Stuart Fork Creek	SFK	43	0.46	0.52	10.90	8.99	0.07
	Swift Creek	SWFT	46	0.45	0.50	10.30	8.85	0.00
	Above Trinity Lake	TRL	37	0.45	0.50	11.30	9.71	0.13
	East Fork Trinity	EFT	40	0.50	0.56	11.40	9.52	0.13
	Mean		**41**	**0.49**	**0.54**	**11.07**	**9.46**	**0.07**

Table 5. Mean pairwise standardized F_{ST} between the Klamath, Trinity, and Jenny Creek (JEN) for Klamath-Trinity Basin speckled dace. Numbers in italics along the diagonal are within basin values.

	Klamath	Trinity	Jenny Creek
Klamath	*0.052*		
Trinity	0.480	*0.070*	
Jenny Creek	0.521	0.606	*n/a*

However, geographic distance only explained 14% of the variation in genetic differentiation between populations in the Klamath (R^2 = 0.145, P=0.0115), whereas 57% of the variation in genetic differentiation was explained for the Trinity (R^2 = 0.581, P= 0.0005). The slope of Trinity IBD relationship (6.350e-04) was steeper than the Klamath IBD relationship (3.550e-04). Jenny Creek (JEN) was excluded for tests of IBD because of its deep level of divergence from all populations.

The unrooted neighbor-joining tree revealed two distinct groups of speckled dace and an intermediate population between them (Fig. 2). One group included all of the populations from the Klamath, except for the Salmon River (NFS and SFS), which were resolved as genetically similar to the Trinity group (hereafter grouped with the Trinity for microsatellite analysis). The second group included all of the populations from the Trinity except for Tish Tang Creek (TT), which was intermediate between the Klamath and Trinity groups. Within the Klamath group, there was some additional structuring of upper Klamath populations (SPE with LNK and SEV with SPR). Jenny Creek (JEN) was located just outside of the main clustering of the Klamath but was highly divergent from other populations in the group.

The DAPC analysis revealed that the first 98 principal components of the PCA captured 90% of the total genetic variation and were retained for the DA (Fig 3). Ordination of the first two axes of DAPC indicated that populations from the Klamath clustered together, while Trinity populations clustered together separately from the Klamath. Jenny Creek was resolved to be highly divergent from other populations and was therefore removed from this analysis to improve the resolution between the Klamath and Trinity clusters (Appendix C).

Bayesian cluster analysis using STRUCTURE suggested that the data was best described by two genetic clusters (K=2) using the *ad hoc* statistic ΔK (Fig. 4, Appendix C). In the structure bar plot, Klamath populations assigned to one group and the Trinity populations assigned to the other group with little admixture observed. Inspection of the distribution of individual assignments at K>2 indicated additional structure. At K=3, Jenny Creek (JEN) was a distinct cluster and at K=4 additional sub-structure was observed in the Upper Klamath (SPE, LNK, SPR, and SEV) and the Salmon River Populations (NFS and SFS).

Hybridization

The distribution of individual admixture proportions (q) from STRUCTURE indicated most individuals assigned to either pure Trinity ($q > 0.9$) or pure Klamath ($q < 0.1$) with 10% of individuals (n=109) being assigned as putative hybrids ($0.1 < q < 0.9$) (Fig 6 and 7). The population with the highest levels of hybridization was Tish Tang Creek (TT), with 48% of its individuals (n=19) assigned as putative hybrids. To a lesser extent, the Salmon River populations (NFS and SFS) showed evidence of hybridization

with 23% of individuals (n=18) assigned as putative hybrids (Fig. 8). However, hybrids were not assigned with high confidence, as no individuals from Tish Tang (TT) and only three individuals from the Salmon (NFS and SFS) had 90% probability intervals that were entirely contained with 0.1 and 0.9 (Fig. 6 and 8).

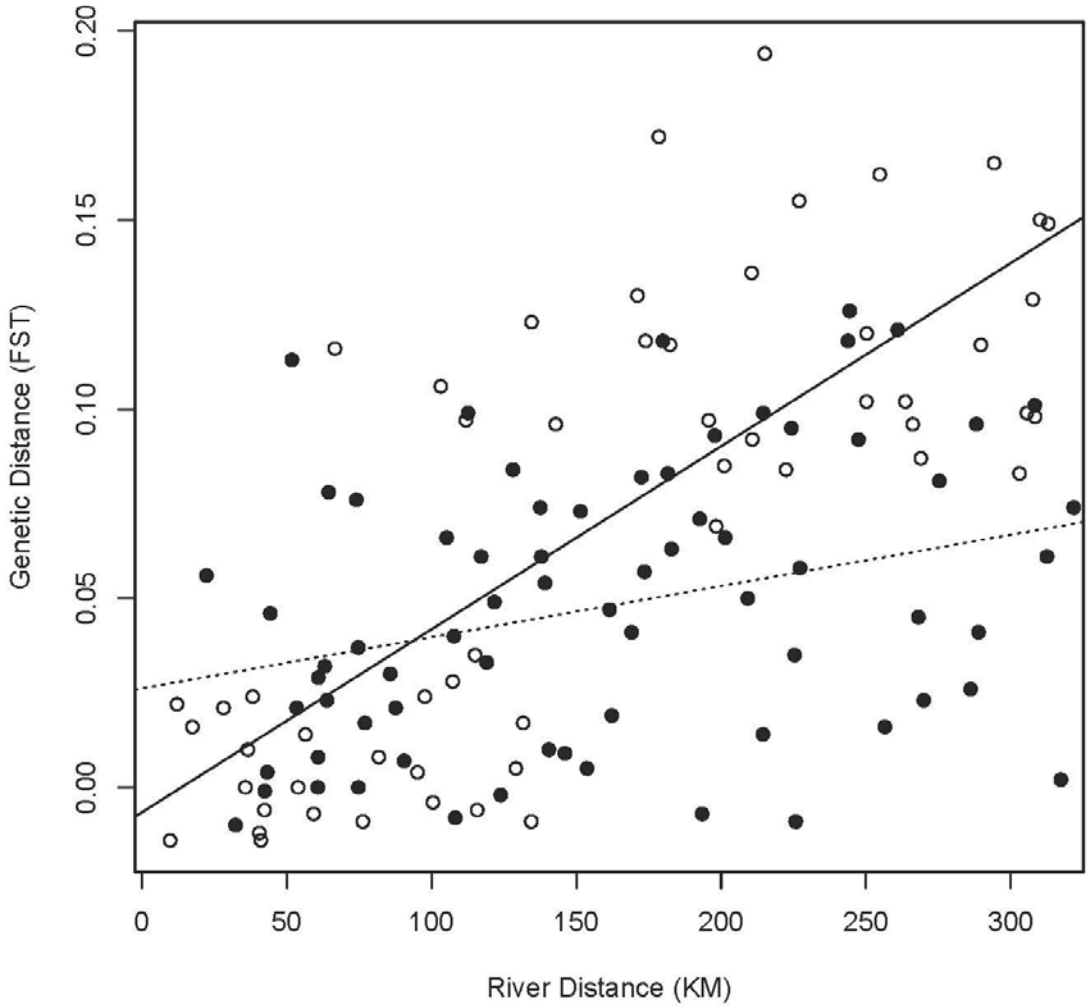

Figure 2. Relationship of pairwise genetic distance (standardized F_{ST}) and river distance (kilometers) for Klamath (solid spots and dotted line) and Trinity (open spots and solid line). Jenny Creek (JEN) was removed from the analysis.

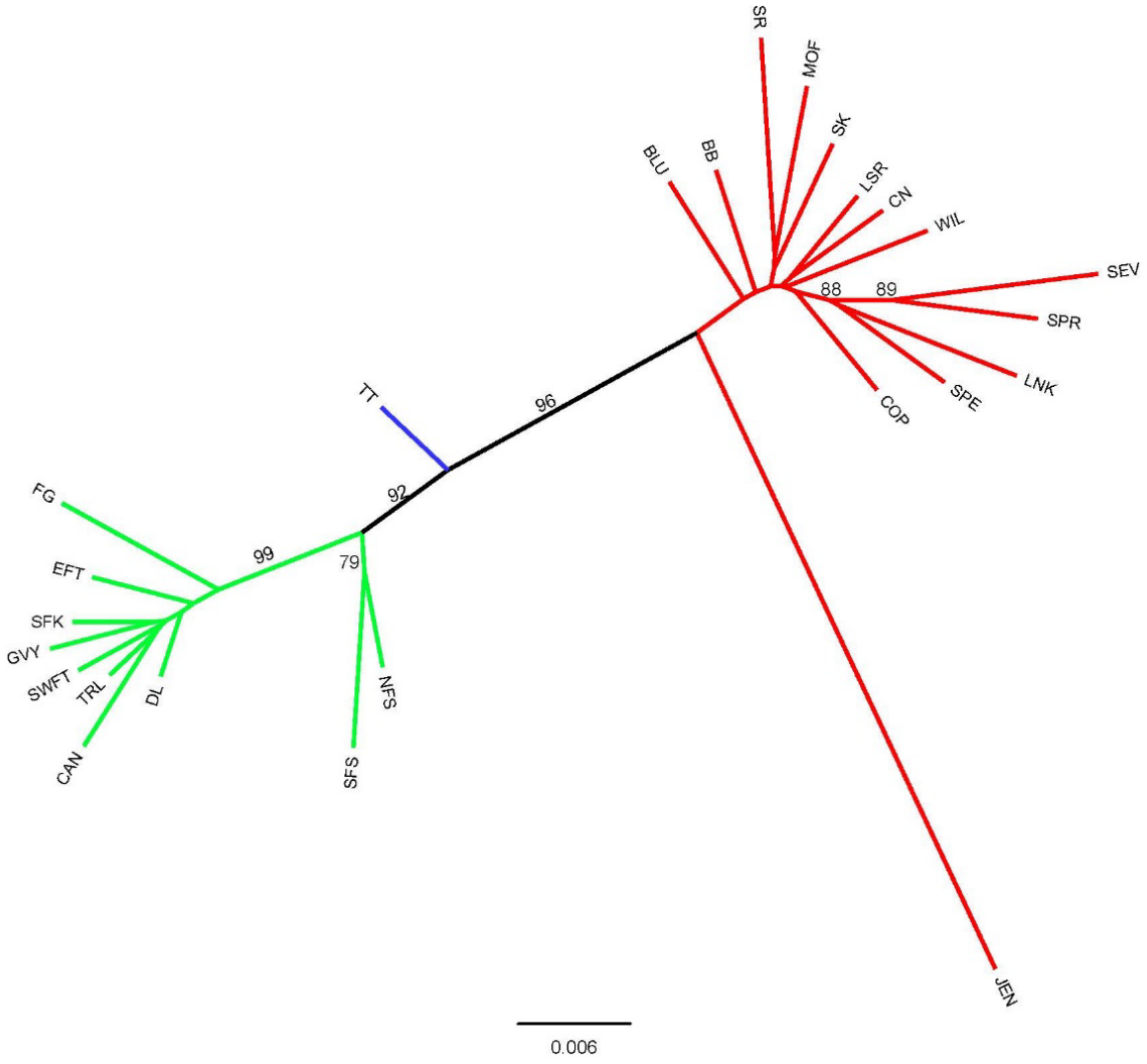

Figure 3. Unrooted neighbor-joining tree based on the Cavalli-Svorza and Edwards's chord distances calculated using the microsatellite data for Klamath-Trinity speckled dace. The numbers on the branches are bootstrap values from 1,000 replicates. Only values above 75 are shown. Trinity branches are green, including the Salmon River populations (NFS and SFS), Klamath branches are red and Tish Tang Creek (TT) is blue.

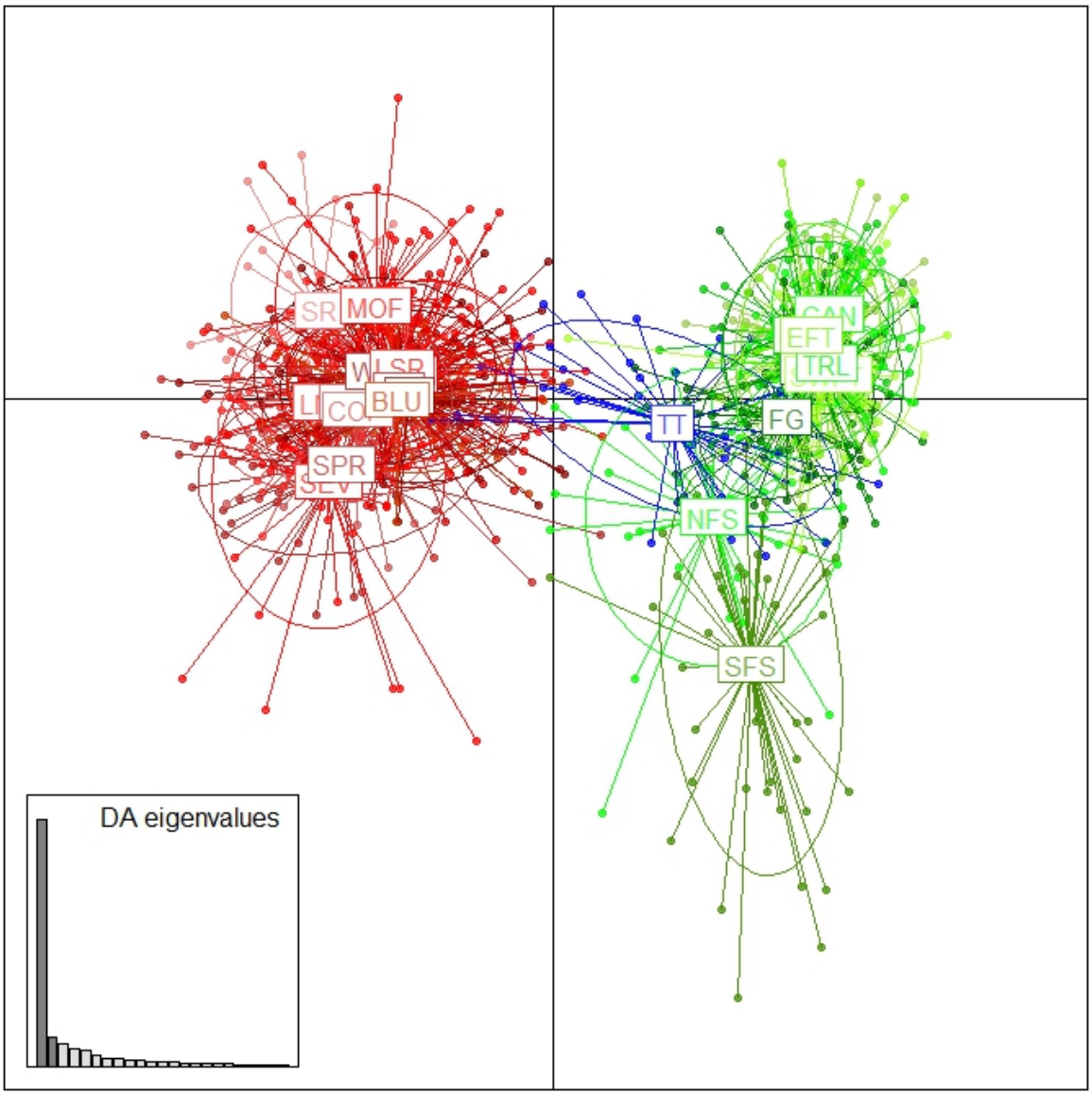

Figure 4. DAPC scatter plot showing 24 Klamath-Trinity speckled dace populations (Jenny Creek removed) using population locations as priors. Populations are labeled inside their 95% inertia ellipses and the dots radiating out are individuals. Klamath populations are shades of red, Trinity populations are shades of Green and Tish Tang Creek (TT) is blue.

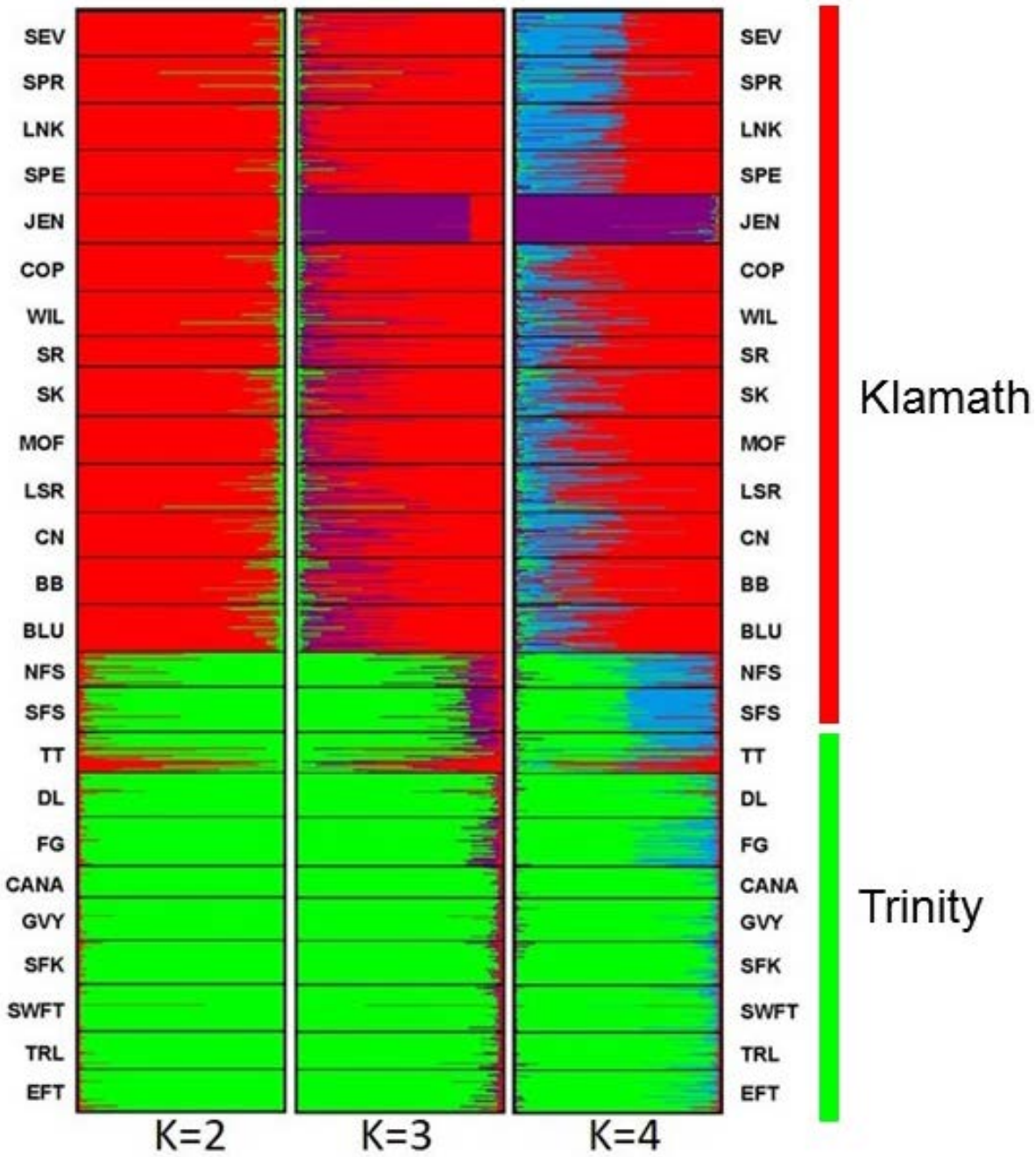

Figure 5 . Results from Bayesian cluster analysis using STRUCTURE assuming two, three, or four genetic clusters (K=2, K=3, K=4) for Klamath-Trinity Basin speckled dace. Each individual is displayed as a thin horizontal line divided into sections, whose length is equal to the probability of membership to a cluster (*q*) while populations are differentiated by thick black lines. K=2 was the most probable cluster based on *ad hoc* statistic ΔK (Appendix C).

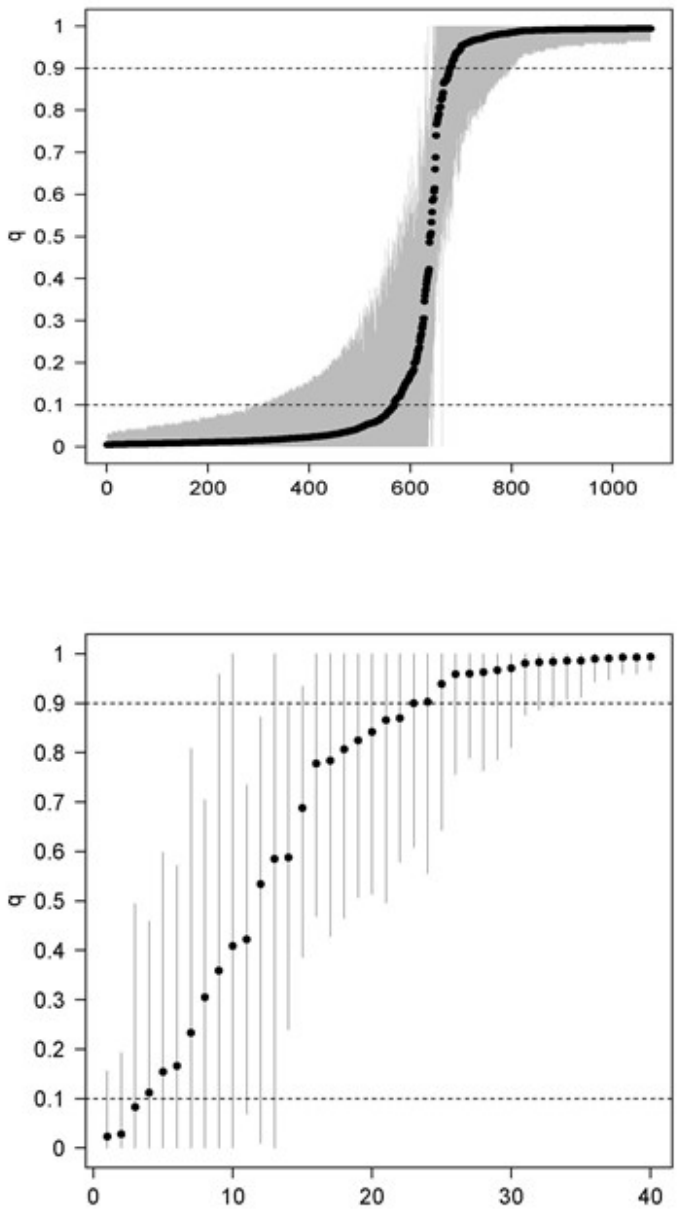

Figure 6. Individual admixture proportions (*q*) estimated by STRUCTURE (K=2) and 90% probability intervals for Klamath-Trinity Basin speckled dace. (a) All individuals in the data set, and (b) Tish Tang Creek (TT). Individual *q* values close to zero and one signify Klamath and Trinity clusters, respectively. Individual *q*-values are sorted from low to high for presentation.

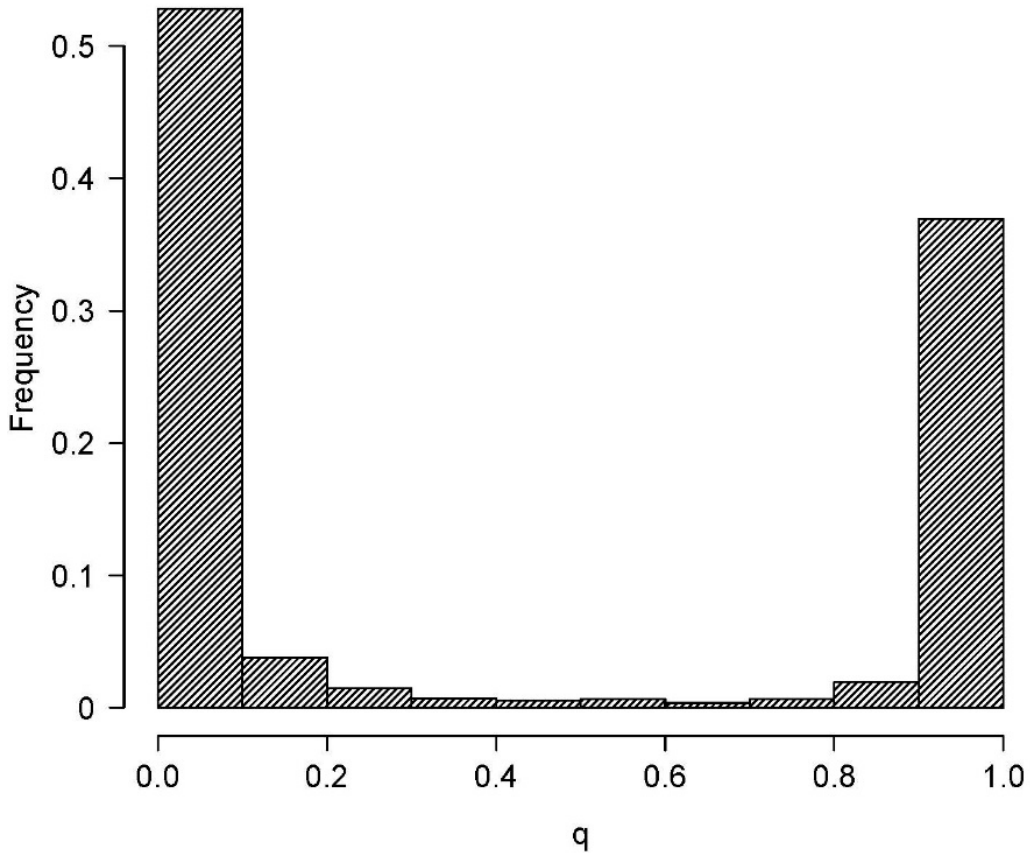

Figure 7. Frequency distribution of individual admixture proportions (*q*) estimated by STRUCTURE (K=2) for Trinity-Klamath Basin speckled dace. Admixture proportions near zero are the Klamath cluster, while *q* values near 1 are the Trinity cluster.

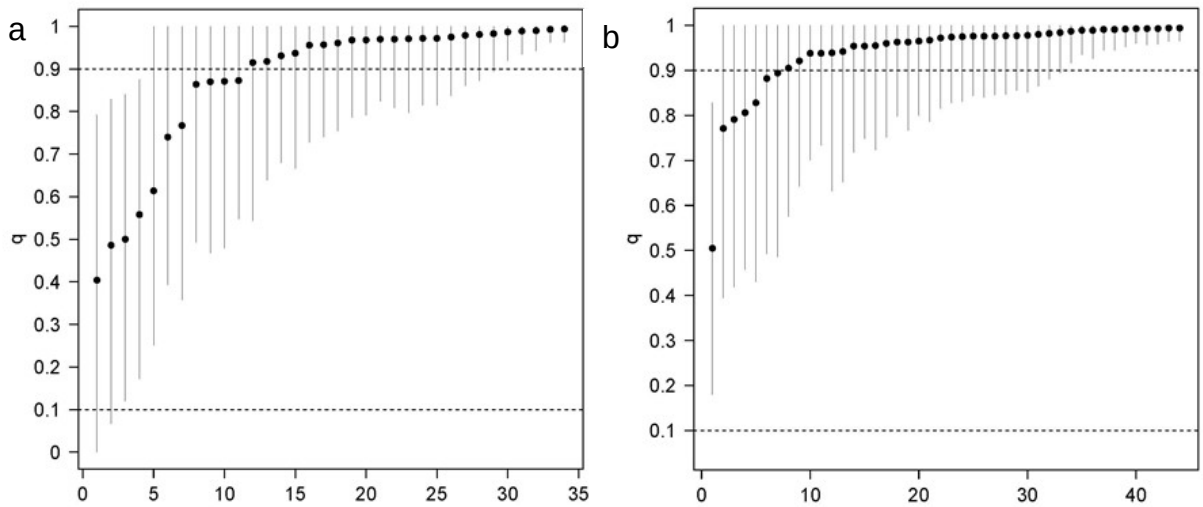

Figure 8. Individual admixture proportions (*q*) estimated by STRUCTURE (K=2) and 90% probability intervals for speckled dace from (a) North Fork Salmon (NFS) and (b) South Fork Salmon. Individuals with *q* values close to zero and one are Klamath and Trinity clusters, respectively.

Mitochondrial Results

For mtDNA, the average number of fish sequenced per location was 9 and ranged from 2-23 (Table 5). A total of 78 haplotypes were defined by 87 variable nucleotide positions. Mean nucleotide diversity was 1.8% in the entire mtDNA dataset. Within populations, the number of haplotypes ranged from 2-9 and nucleotide diversity ranged from 0.036-3.1%.

The Klamath displayed higher mtDNA diversity than the Trinity (Table 5). Unlike the microsatellite results, the Salmon River (NFS and SFS) haplotypes were most related to Klamath populations and were included with the Klamath for mtDNA diversity calculation and discussion. Klamath Basin populations contained 55 haplotypes (n=137) with a haplotype diversity of 0.949, while the Trinity River populations contained 28 haplotypes (n= 99) with a haplotype diversity of 0.665. Nucleotide diversity was also found to be lower in Trinity populations (0.67%) compared to Klamath populations (1.4%), though, nucleotide diversity within the Klamath was inflated by a single highly divergent haplotype (Sev1). The Salmon River had reduced levels of mtDNA diversity, especially in the South Fork Salmon River (SFS).

Table 6. Mitochondrial DNA sequence diversity for Klamath-Trinity speckled dace including river system, population abbreviation (ID), number of sequences (n), number of variable sites in the sequences (S), sequence diversity (π), unique haplotypes (H), and haplotype diversity (Hd).

River System	ID	N	S	π	H	Hd
Klamath						
	SEV	6	40	0.031	4	0.869
	SPR	10	13	0.007	7	0.911
	LNK	7	25	0.018	7	1.000
	SPE	9	23	0.012	7	0.944
	JEN	12	4	0.002	4	0.561
	COP	8	11	0.007	6	0.893
	WIL	8	11	0.009	6	0.893
	SR	16	16	0.008	9	0.817
	SK	8	9	0.005	7	0.964
	MOF	8	9	0.008	3	0.607
	LSR	7	13	0.009	7	1.000
	CN	7	10	0.006	4	0.714
	BB	2	7	0.013	2	1.000
	BLU	8	12	0.009	6	0.929
	NFS	11	4	0.002	4	0.600
	SFS	10	1	0.000	2	0.200
	Mean	**9**	**13**	**0.009**	**5.3**	**0.806**
Trinity						
	TT	15	26	0.018	9	0.905
	DL	8	4	0.002	5	0.786
	FG	15	7	0.003	7	0.857
	CAN	23	5	0.001	6	0.518
	GVY	7	4	0.002	4	0.714
	SFK	8	1	0.000	2	0.250
	SWFT	8	3	0.001	3	0.464
	TRL	7	1	0.001	2	0.286
	EFT	8	2	0.001	3	0.464
	Mean	**11**	**6**	**0.003**	**4.6**	**0.583**

ML Tree

The Klamath-Trinity Basin speckled dace were resolved as monophyletic (bootstrap (BS) 77), excluding one highly divergent haplotype, and exhibited a sister group relationship with nearby basins in California and Oregon (Sacramento, and Pit Rivers with the Goose Lake Basin in Oregon, Fig. 9)). The exception to monophyly was a haplotype identified in a single individual from a creek in the upper Klamath River (Sev1). This unique haplotype was highly divergent from all other Klamath-Trinity Basin haplotypes (sequence divergence > 5.5%, Table 8). A BLAST search (Altschul et al. 1990) for similar sequences in GenBank revealed that Sev1 was most similar (99%) to three Upper Klamath haplotypes (AY366268, AY366269, AY366279) identified by Pfrender et al. (2004) as Klamath B. Next most similar (96%) were speckled dace from the Lahontan Basin in Nevada (Billman et al. 2010, Houston et al. 2012).

Within the Klamath-Trinity Basin three clades were identified: (i) Trinity (BS = 73), (ii) Klamath I (BS = 89), and (iii) Klamath II (BS = 65) (Fig. 7). The Trinity was composed exclusively of individuals (n=71) that originated from the Trinity River. Klamath I included the majority of the individuals collected from Klamath River (n=118), and 10 of the 15 speckled dace from the lower Trinity River tributary Tish Tang Creek (TT). Klamath II was composed of 18 individuals restricted to sites in the upper Klamath River of Oregon including all 12 individuals from Jenny Creek (JEN). The Trinity clade was resolved as sister to Klamath II (BS = 89) and Klamath I was sister to this group. The relationship of Klamath II clade, consisting of Upper Klamath haplotypes

that are closer related to the Trinity clade than the rest of the Klamath, is puzzling given the distances involved and may reflect incomplete lineage sorting. Klamath I contained considerably more within-group structuring than the Trinity clade, including a well-supported (BS= 88) branch containing haplotypes from above Upper Klamath Lake in Oregon.

In the 95% maximum parsimony network Klamath-Trinity Basin speckled dace were divided into two networks separated by 12 substitutions (Fig. 10). The same three clades resolved by the ML tree were evident: (i) Trinity, (ii) Klamath I, and (iii) Klamath II (Figure 8). Klamath I displayed considerable structure in the haplotype network and possessed the largest within-group percent sequence divergence (0.874, Table 7). The Trinity clade had the lowest within group percent sequence divergence (0.141) containing one primary haplotype occurring at high frequency in all populations and then additional haplotypes radiating out from it. Klamath I and Trinity displayed the largest sequence divergence (2.96%), followed by Klamath I versus Klamath II (2.72%), and lowest was Trinity versus Klamath II (1.29%).

Table 7. Percent sequence divergence of mitochondrial cyt b sequences D_{xy} between groups as defined by the haplotype network. Percentages along the diagonal in italics are within group diversity, Sev1 was only collected from one individual.

	Klamath I	Trinity	Klamath II	Sev1
Klamath I	*0.874*			
Trinity	2.96	*0.141*		
Klamath II	2.72	1.29	*0.36*	
Sev1	5.93	5.68	5.95	*N/A*

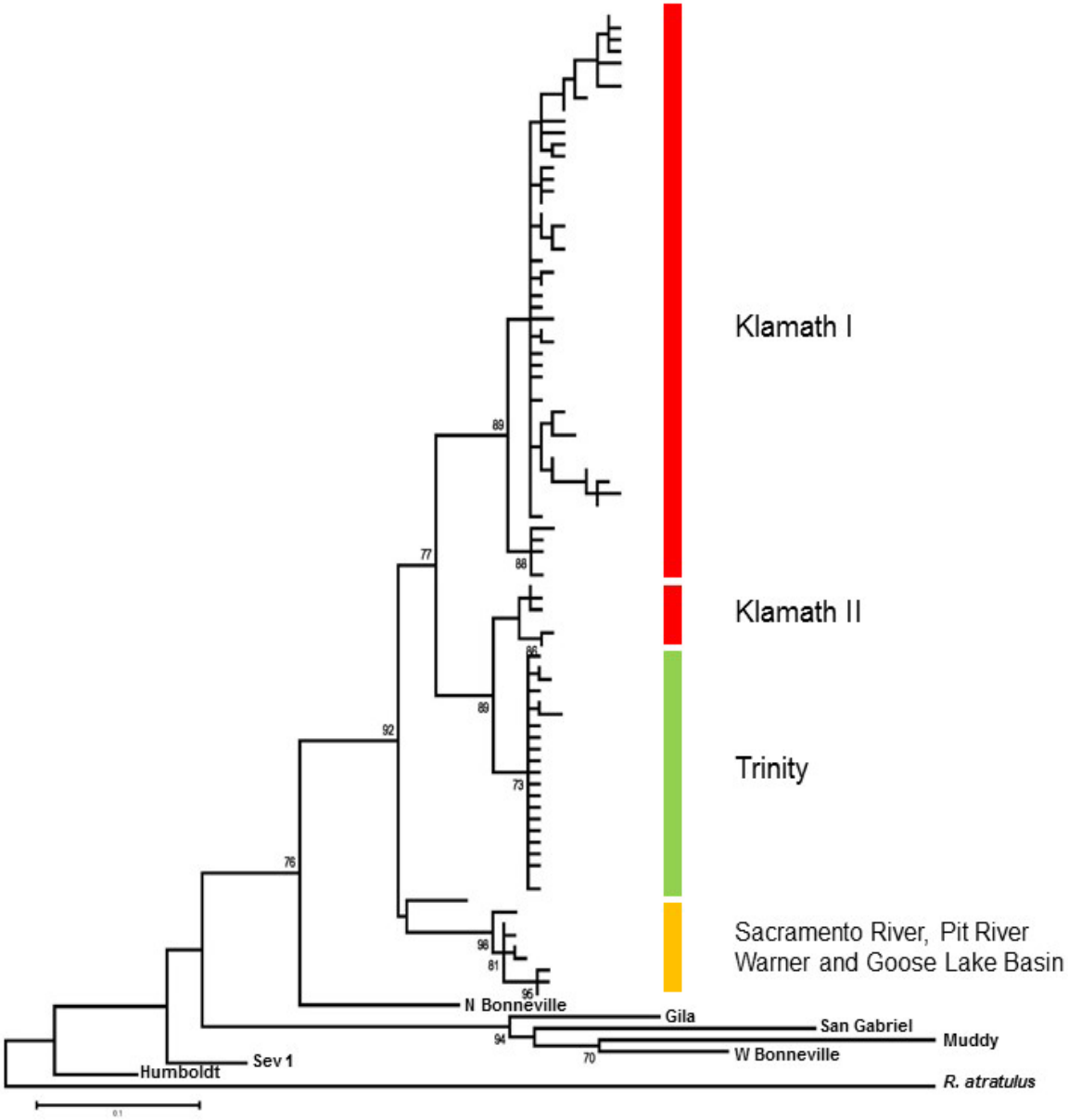

Figure 9. Maximum Likelihood tree generated from unique mtDNA cyt *b* haplotypes for Klamath-Trinity speckled dace and. A single *R. atratulus* sequence served as an out-group. The ML tree resolved four clades: Klamath I, Trinity, Klamath II, and a clade consisting of: the Sacramento and Pit River with Warner and Goose Lake Basin from Oregon. Haplotypes from more distant basins with the divergent haplotype (Sev1) are basal to the four clades. Support for the tree was established via 1000 bootstrap replicates, only values above 70 are shown.

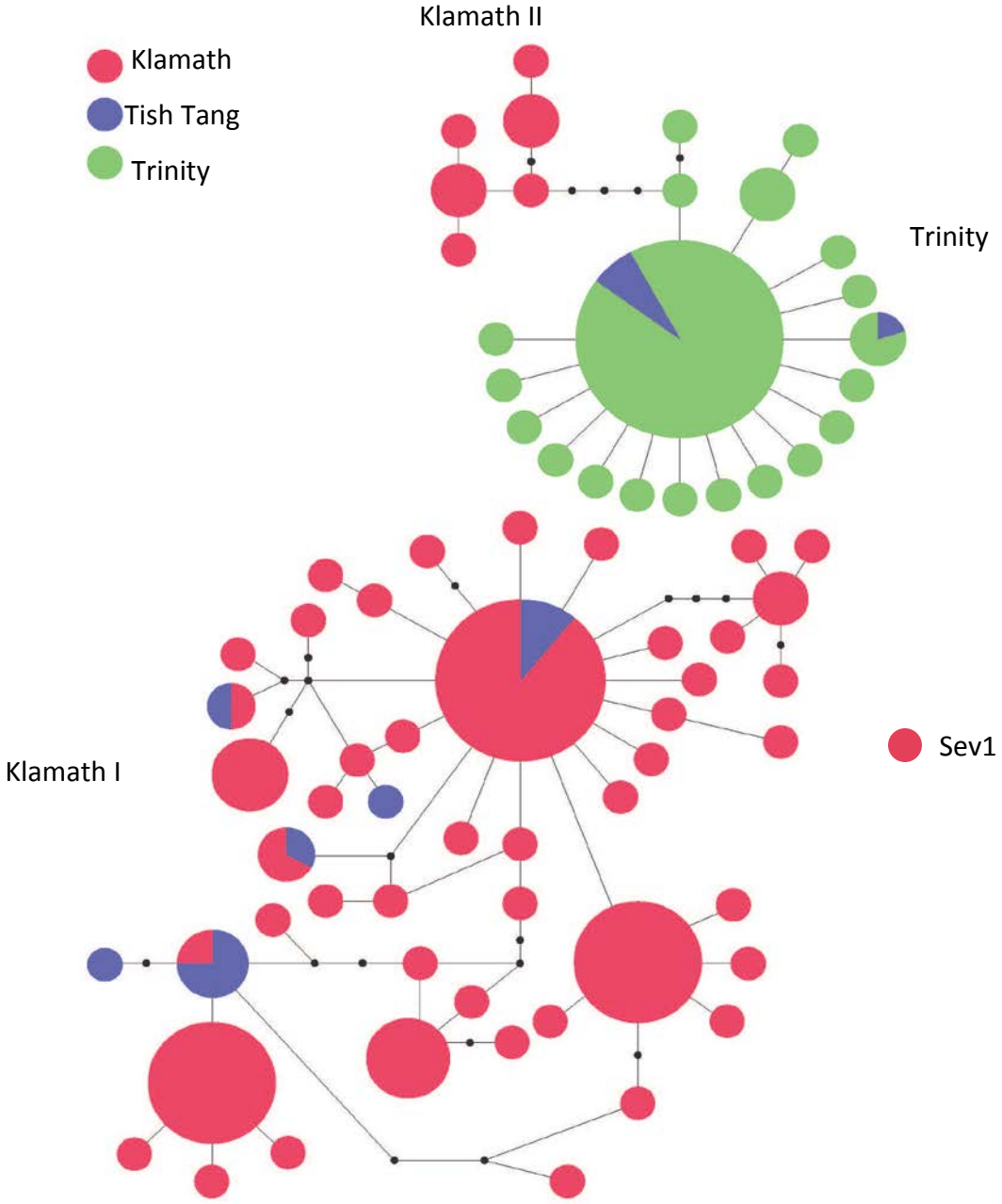

Figure 10. Parsimony haplotype network (95%) based on mtDNA cyt *b* sequences for Klamath-Trinity Basin speckled dace. Unique haplotypes are represented by circles, the size of the circle corresponds to haplotype frequency. Small black dots represent inferred haplotypes. Three groups were resolved: Klamath I, Trinity and Klamath II as well as the divergent haplotype Sev1. Haplotypes from Klamath populations are red, Trinity populations are green, and Tish Tang Creek (TT) is blue.

DISCUSSION

This research provides the first comprehensive study of genetic variation within Klamath speckled dace. A robust set of collections from throughout the geographic range of the subspecies was assayed using both mtDNA and microsatellite loci and uncovered several interesting patterns. First, speckled dace in the Klamath-Trinity basin form a nearly monophyletic group that is genetically divergent from other geographic regions. Second, the Klamath-Trinity River is inhabited by two divergent lineages of speckled dace, one cryptic lineage confined to the Trinity River system and the main lineage occurring throughout the remainder of the Klamath River system. Third, the two lineages contact each other near the confluence of the Klamath and Trinity Rivers but hybridization between them appears to be rare. Fourth, within basin patterns of genetic diversity and structure differed between the Klamath and Trinity lineages, and the Salmon River populations (NFS and SFS) displayed disagreement in their microsatellite and mtDNA assignment. Finally, speckled dace occurring in the upper Klamath appeared to have a complex evolutionary history, as indicated by the detection of a divergent population in Jenny Creek (JEN), and the presence of a highly divergent haplotype (Sev1).

Genetic Divergence

The most significant result of this study was the deep genetic break between speckled dace from the Klamath and Trinity. Divergence was resolved using both nuclear microsatellites and mtDNA sequence data and using variety of analysis methods (trees, multivariate, and Bayesian). This pattern of divergence between the lineages was complicated by the Salmon River (NFS and SFS), a mid-basin Klamath tributary, which resolved as a Klamath population according to mtDNA analysis and as Trinity in the nuclear microsatellite analysis. This discordance between the microsatellites and mtDNA haplotypes is most likely the product of hybridization and highlights the importance of using genetic multiple markers to infer evolutionary history.

The level of genetic divergence in the mtDNA cyt *b* between Klamath and Trinity lineages (~3%) was similar to mtDNA divergence resolved in sibling fish species (~2%; Ward et al 1994). Using Smith and Dowling's (2008) estimate of cyt *b* mutation rate of 1.2% to 2.5% per million years for speckled dace as a crude guideline, the approximate date of divergence between the Trinity and Klamath lineages was during the Pleistocene, 1.18 to 2.47 million years ago. The extent of mtDNA genetic divergence between the Klamath and Trinity lineages is similar to divergence levels recovered between allopatric speckled dace populations in river basins where contemporary gene flow is seemingly impossible. For example, speckled dace in Warner and Goose Lake Basins in Oregon had similar cyt *b* nucleotide diversity between samples (3.0-3.6% D_{xy}, Arden et al. 2010). While high levels of genetic differentiation between basins generally occurs in speckled

dace (Oakey et al 2004; Pfrender et al 2004), it is noteworthy that the genetic break between Klamath and Trinity lineages occurred despite the absence of known physical barriers. When within basin divergence has been resolved in speckled dace, it has generally been attributed to a barrier, such as the Grand Canyon (Smith and Dowling 2008) or a historic drainage rearrangement (Oakey et al. 2004).

The close relationship between the clade containing speckled dace from the Sacramento, Pit, Goose Lake and Warner Basin with the Klamath-Trinity Basin clade (3.6% sequence divergence) reflect the historic connections these drainages had with the hypothesized westward flowing Proto-Snake River (Oakey et al 2004; Arden et al. 2010). Historically, Goose lake overflowed and spilled into the Pit River (Baldwin 1981), explaining the association these Oregon Great basin drainages have with the Pit and Sacramento rivers.

The deep divergence between Trinity and Klamath lineages, combined with the absence of examples sympatric speckled dace lineages in other areas, suggests that divergence occurred in allopatry and more recent geological processes have resulted in secondary contact. Additionally, the lack of shared haplotypes between the lineages, except at the confluence, suggests the river systems were historically separated (Blakney et al. 2014). Unfortunately, the precise biogeographic processes in the Klamath-Trinity Basin that have impacted speckled dace populations during the Pliocence-Pleistocene have been obscured by repeated uplift over short time-scales (Aalto 2006). Furthermore, repeated glaciation advance and retreat and erosion make reconstruction of historical river drainage patterns uncertain (Anderson 2008). The evidence that remains suggests

the northwest Klamath Mountains are the result of recent uplift during Pliocene-Pleistocene and before that, the area was a low-lying peneplain (Aalto 2006). Before this uplift, the Trinity and Klamath Rivers presumably had different drainage patterns then they do today, though, the nature of their flow and outlet(s) are obscured (Anderson 2008). Thus, it is reasonable that the Klamath and Trinity Rivers may have historically been separate systems allowing for allopatric diversification. The estimated divergence time for Klamath and Trinity lineages (1.18 to 2.47 million years ago) is consistent with the estimated time of uplift for the northwest Klamath Mountains (Aalto 2006).

Hybridization

Analysis of nuclear microsatellites strongly suggested that most individual speckled dace were either pure Klamath or pure Trinity with very few hybrids between them. The detection of few hybrids between the Klamath and Trinity lineages is consistent with other studies of parapatric cryptic species (Brown et al. 2007). The mtDNA analysis confirmed the lack of hybridization by evidence of only one population sharing haplotypes from both river systems. Hybridization was predominately confined around the confluence of the rivers at Tish Tang Creek (TT) or in a Klamath mid-basin tributary, the Salmon River (NFS and SFS). Based on the differences in marker agreement/disagreement detected in Tish Tang Creek (TT) and Salmon River (NFS and SFS), hybridization may have taken place at different time frames (Koblmüller et al. 2010) or the modes of contact differed (Currant et al. 2008). In both sites, shifts in historical distributions and range expansions forced by historical processes most likely

caused secondary contact between the lineages. Alternatively, hybridization levels may be higher than detected (especially back crossed hybrids) but the resolution of microsatellite markers may not have been sufficient to accurately identify them (Boecklen and Howard 1997; Vähä and Primmer 2006).

Salmon River Hybridization

Speckled dace in the Salmon River (NFS and SFS) exhibited discordant patterns in mtDNA and nuclear microsatellites. All Salmon River individuals had Klamath mtDNA haplotypes (n=21), whereas nuclear microsatellite analysis placed the Salmon River with the Trinity lineage with some Klamath admixture (Fig. 5 and 7). But, there was agreement between the two markers on the overall low genetic diversity contained in the Salmon River, especially in the South Fork Salmon River (SFS). The reduced genetic diversity, coupled with the paraphyly of the genetic markers observed in Salmon River speckled dace, is characteristic of bottleneck stemming from a founder event followed by introgression (Currat et al. 2008, Dlugosch and Parker 2008).

The lack of hybrids in the Klamath River around the Salmon River confluence (BB and CN) suggests that Trinity speckled dace are not migrating through the Klamath River to enter the Salmon River. Contact between Trinity and Klamath lineages within the Salmon River have presumably resulted when the Salmon River captured portions of the upper Trinity River. For example, the upper reaches of Coffee Creek, a Trinity River tributary, are known to have been captured by the South Fork of the Salmon River during the Pleistocene (Hershey 1900; Sharp 1960). Expansion into the Salmon River might

have been enabled by similar habitat characteristics (clear cold water and steep gradient streams) shared by the upper Trinity tributaries and Salmon River.

If a species geographically expands its range into a closely related species' range and hybridization occurs, asymmetrical mtDNA introgression may occur (Keck and Near 2010; Wielstra and Arntzen 2012). When the relative abundances of species are unbalanced, as they would be for invading Trinity speckled dace, the rarer species is expected to be more likely to mate with conspecifics (Hubbs 1955). This mechanism, called the "desperation hypothesis" or Hubbs principle, has been observed in a number of organisms (Hubbs 1955; Grant and Grant 1997; Lepais et al. 2009). Since mtDNA is inherited maternally and does not dilute as nuclear DNA does, it either gets passed down whole or not at all. Backcrossing of females with Klamath mtDNA with Trinity speckled dace or hybrids would progressively dilute the nuclear DNA, while Klamath mtDNA would remain. Factors such as sex biased dispersal (Berthier et al. 2006), fitness differences between the lineages (Wielstra and Arntzen 2012), or mate choice (Wirtz 1999) could have played a role in female Trinity genes not being inherited. While incomplete lineage sorting could account for the marker discordance observed in the Salmon River (NFS and SFS), the Salmon River's proximity to the Trinity River, coupled with geologic evidence, make it unlikely. In fact, the Salmon River speckled dace may represent an additional discrete lineage in the basin composed of hybrids (a hybrid swarm), based on the region wide pattern of marker discordance (Allendorf et al. 2001).

Trinity Klamath Hybrid Zone

The area around the confluence of the Klamath and Trinity Rivers likely constitutes a hybrid zone, as indicated by the confinement of the Klamath and Trinity lineages to their respective river systems and the hybridization detected near the confluence. Tish Tang Creek (TT) was the only population to share haplotypes from both Klamath and Trinity lineages, suggesting that hybridization is occurring on a contemporary timescale (Keck and Near 2009). In the microsatellite analysis, Tish Tang Creek (TT) was resolved as intermediate in the neighbor joining tree (Fig 3) and the DAPC (Fig 4) and contained the highest number of individuals with a *q* value between 0.1 and 0.9 (though assignments were not confident) (Fig. 6). The increased genetic diversity found within Tish Tang Creek (TT) compared to other Trinity populations is also evidence of hybridization (Peters et al. 2007). A possible hybrid population was also detected less than 15 river kilometers upstream from Tish Tang Creek (TT) in the Trinity River by Kinziger et al. (2011). The contact zone is most likely geographically small, as three populations (BLU, BB, and DL) all less than 80 river kilometers from Tish Tang (TT) exhibited little to no evidence of hybridization.

There are many theoretical models on the structure and location of hybrid zones. The most common model is a tension zone (Barton and Hewitt 1985), where the hybrid zone is defined by dispersal and selection acting upon hybrids, with hybrids displaying lower fitness than parental types (Harrison 1993). Alternatively, hybrid zones can be distributed and maintained based on environment (Gow et al. 2007), genetic

compatibilities (Smith et al. 2003), habitat choice (Via et al. 2000) and mate preference (Vamosi and Scluter 1999).

Hybrid offspring has generally been reported to be less fit in freshwater fishes, compared to parental types (Scribner et al. 2001). However, this is not always the case as naturally occurring hybrid offspring can be an excellent source of evolutionary potential, and able to exploit novel habitats (Arnold and Hodges 1995; Arnold et al. 1999). Given that hybridization appears to be limited to a narrow area where the lineages come into contact, and when dispersal potential of speckled dace is high (Pearsons et al., 1992; Brown and Moyle 1997), this suggests that there is a mechanism that is minimizing hybridization.

It is often difficult to resolve the forces responsible for the location and structure of a hybrid zone, as well as the isolating mechanisms involved (Nielsen et al. 2003). Unlike other cyprinids (reviewed in Scribner et al. 2001), speckled dace hybridization and potential reproductive isolating mechanisms have not been well studied. Hybrid crosses with speckled dace have been reported in longnose dace *(Rhinichthys cataractae)* and redside shiners (*Richardsonius balteatus*), distinguished by morphological characteristics (Smith 1973) but not confirmed by genetic analysis. In a molecular study, introduced speckled dace in the Ruby Valley, NV were found to hybridize with relict dace (*Relictus solitarius*) (Houston et al.2012), but a reproductive isolating mechanism was not proposed.

There are several hypothetical isolating mechanisms that could keep Klamath and Trinity speckled dace lineages distinct. Post zygotic intrinsic barriers, such as genetic

incompatibility or hybrid sterility, are always a possibility (Harrison 1993). Hybrid offspring would only be produced very rarely or have reduced fitness compared to parental types in this scenario. Genetic incompatibility is unlikely to be the only barrier to gene flow between the lineages, as intergeneric hybridization between speckled dace has apparently produced viable offspring (Smith 1973, Houston et al 2012).

Fish species with pre-zygotic isolating mechanisms, such as complex mating rituals seen in Centrarchidae, are expected to have reduced hybridization rates (Scribner et al. 2000). In other *Rhinichthys* species, differences in spawning behavior is one factor that has been reported to keep sympatric longnose dace (*R. cataractae*) and blacknose dace (*R. atratulus*) from hybridizing (Bartnik 1970). Speckled dace reproductive behavior has not been studied comprehensively, but limited field observations suggest that the many males will congregate on a nest site and swarm females when they enter, thus attempting to copulate with them (John 1963; Mueller 1984; O'Brien 2013). This reproductive strategy, if accurately depicted, would limit the effectiveness of behavior as an isolating mechanism between the lineages.

The most plausible isolating mechanism for Klamath and Trinity speckled dace is spatial isolation, where hybridization is restricted by occurring or breeding in different locations (Scribner et al. 2000). Presumably, spatial isolation may isolate slow and fast water forms of speckled dace that have been observed in the Colorado River Basin (Smith and Dowling 2008) and Columbia River Basin (Peden and Hughes 1988). Adult longnose and blacknose dace were also specially isolated by water velocity and spawning substrate preference (Bartnik 1970). While the habitats of the Lower Klamath and the

Upper Klamath Basin are dissimilar, the Trinity River system is differentiated from the rest of the Lower Klamath by higher elevation, steeper gradient, and cooler spring flows due snowmelt (Jhingran 1948, United States Department of Agriculture 2005).

Trinity and Klamath lineages may be similar to the previously mentioned swift and slow water forms of speckled dace, adapted to the habitat conditions of their native river systems (Peden and Hughes 1988; Smith and Dowling 2008). Temperature differences between the Klamath and Trinity River systems during the summer when speckled dace spawn (Moyle 2002) could have played a role in temporally isolating the lineages. The mean monthly temperature in the Lower Klamath is 20-24°C for July and August, with little impact from dams due to their distance (Bartholow 2005). In the Upper Trinity Basin at Lewiston, mean monthly temperatures range from 16-20°C during the summer, before dam construction. In natural observations (John 1963, O'Brian 2013) and laboratory studies (Kaya et al. 1991), water temperature, in addition to photoperiod and rising water level, were documented to affect spawning time in speckled dace. Specifically in the Trinity River, rising water temperatures were cited as a factor that most likely triggered spawning in speckled dace (Jhingran 1948). Conflicting or only partially overlapping spawning times caused by differences in water temperatures between the Trinity and Klamath River systems could act as a powerful isolating mechanism. A combination of isolating mechanisms working in conjunction is the most likely explanation for the lack of hybridization observed between the lineages, as Bartnik (1970) observed for blacknose and longnose dace.

Within Basin Diversity

As a whole, speckled dace in the Klamath-Trinity Basin contained high levels of genetic diversity, with more diversity generally observed in the Klamath than in the Trinity (Tables 4 and 6). In a comparison to other basins, Pfrender et al. (2004) found that speckled dace in the upper Klamath had the highest within-basin diversity.

Trinity

The entire Trinity lineage, with the exception of Tish Tang Creek (TT), showed reduced levels of genetic diversity in comparison to the Klamath, as well as speckled dace from other areas (Kinziger et al. 2011). This lineage-wide reduction in genetic diversity, observed across both microsatellites (Trinity H_e=0.49, Ar=9.46, Klamath H_e=064, Ar=12.35) and mtDNA (Trinity Hd=0.583, Klamath Hd=0.806), is presumably a consequence of a historic demographic bottleneck.

One possible cause of a bottleneck in Trinity speckled dace is a number glaciation advances and retreats, which is known to have occurred in the region. While most of the Klamath Mountains remained relatively free of glaciation during the coldest parts of the Pleistocene (Orr and Orr 2006), the Trinity and Salmon Mountains experienced at least four separate glaciation events during that time (Hershey 1903; Sharp 1960). The Pleistocene glaciation cycles have been one the most substantial events impacting the genetic structure of North American freshwater fishes (Bernatchez and Wilson 1998). While the Trinity Mountains were not covered with sheets of ice seen in more northern latitudes, scattered glaciers in the river system could have fragmented and reduced

Trinity speckled dace populations. Additionally, geologic events, such as the uplift of the Klamath Mountains or drainage rearrangements in the Lower Klamath and Trinity, could have played a role in fragmenting and reducing genetic diversity in the Trinity lineage. Additionally, lower levels of genetic diversity have been associated with steep gradients and an increase in elevation for freshwater fish (Hernandez-Martich and Smith 1990; Castric et al. 2001). The Trinity Mountain range is the highest of the Klamath Mountains and the Upper Trinity Basin is characterized by extremely steep terrain (United States Department of Agriculture 2005). Steep gradients and high elevations are also expected to increase the chance of physical barriers to gene flow (i.e. waterfalls), isolating populations and increasing the effect of drift (Castric et al. 2001), which may explain the stronger IBD relationship observed in the Trinity.

At a more contemporary time-scale, extensive mining on the Trinity River (reviewed in Krause et al 2010) could have further reduced genetic diversity in the Trinity speckled dace. Large scale hydraulic mining started on the Trinity River in the 1860's and lasted 80 years before being replaced by smaller scale dredger mining that operated until the 1960's. These operations diverted water and produced millions of cubic yards of sediment, which must have had a serious impact on fish in the Trinity (Krause et al 2010). Although, the extent of Trinity River mining's impact on speckled dace is unknown and unlikely the sole cause of the lineage wide pattern of reduced genetic diversity.

Klamath

The Klamath, with the exception of the Salmon River (NFS and SFS) and Jenny Creek (JEN), displayed very high levels of genetic diversity spread across the basin. While substantial structure was observed throughout the Klamath, there was very little geographical basis for it, as haplotypes were frequently shared by upper and lower basin populations despite their dissimilarity in habitat and great distance (Appendix E). Although, both microsatellite loci and mtDNA recovered some minor sub-structuring of the Upper Klamath populations (SEV, SPR, LNK, and SPE) that was well supported by the neighbor-joining tree (Fig 2) and the mtDNA ML tree (Fig 6). The Klamath displayed a weak relationship (but significant) of IBD and smaller within river system genetic differentiation compared to the Trinity, in spite of the greater distances in the Klamath (Table 5).

Jenny Creek (JEN), an upper Klamath population, exhibited reduced genetic diversity compared to other Klamath populations and was highly divergent. A natural 10-m waterfall found 4 km from the mouth of Jenny Creek blocks upstream migration of fish and is the most likely explanation for the divergence and low genetic diversity. A federally recognized special status population of Klamath smallscale sucker (*Catostomus rimiculus*) found in Jenny Creek also reflects this isolation (Rossa and Parker 2007). In the microsatellite neighbor-joining tree and DAPC plot, Jenny Creek (JEN) was found to be a highly divergent Klamath population (Fig. 2, Appendix C). While in the mtDNA analysis Jenny Creek (JEN) was surprisingly found to be more similar to the Trinity haplotypes within the Klamath II clade than the main Klamath lineage. Incomplete

lineage sorting of mtDNA in Klamath II seems the most likely explanation for the discordance between markers, possibly caused by rapid expansion and radiation of speckled dace during the Pleistocene or Pliocene. For recently diverged lineages, incomplete lineage sorting in gene trees is a widespread problem for correctly inferring evolutionary relationships (Carstens and Knowles 2007; Waters et al 2010). The geographic distances and physical barriers make hybridization with the Trinity an unlikely explanation for the Klamath II clade. Pfrender et al. (2004) also found Jenny Creek haplotypes to be distinct, though nuclear markers were not employed, and the study was limited to the upper Klamath so the irregularity and similarity to the Trinity clade was not resolved. This highlights the importance of using multiple markers and a broad sampling regime to infer accurate evolutionary relationships (Maddision and Knowles 2006).

The one individual (Sev1) from the Upper Klamath that was highly divergent from all groups in mtDNA (~6%) appeared to be the same divergent lineage detected by Pfrender et al.(2004) based on the blast search. There was no evidence of this divergent lineage in the microsatellite analysis and Sev1 possessed a typical Klamath genotype, suggesting that this divergence is limited to the mtDNA. Alternatively, samples sizes in the upper Klamath may have been insufficient to detect it.

This divergent lineage in the Upper Klamath Basin most likely arrived from historical drainage connections with other basins (Pfrender et al 2004; Arden et al. 2010). Based on the blast search, this divergent lineage has its closest affinities to the speckled dace from the Lahontan Basin. Geologic evidence supports the dispersal of Lahontan

dace into the Klamath Basin through a connection to the pre-modern Snake River that flowed west (Taylor 1985; Minckly et al 1986). The Snake River is thought to have periodic connections with the Lahontan basin and the Upper Klamath Basin (Taylor 1985; Billman et al. 2010), so the Snake River may have acted as a transport from the Lahontan to the Klamath Basin. The similarity in a number of fish species between the Great Basin and the Upper Klamath, as well as fossil evidence, support a Klamath, Snake River, and Great Basin connection as well (Minckly et al. 1986). Pfrender et al. (2004) proposed alternative hypotheses to drainage realignment as explanations for this divergent lineage: (i) incomplete lineage sorting and (ii) the retention of ancestral polymorphisms. However, the similarity of Sev1 to Lahontan sequences and geologic evidence point to drainage rearrangement as the most likely source.

Conclusion and Conservation Implications

The most striking result from this work is the detection of two deeply diverged lineages of speckled dace that show evidence of reproductive isolation within the Klamath-Trinity Basin. Historical events that are responsible for the formation of the cryptic diversity observed in the Trinity are not well understood and show the need for a more comprehensive geologic research of the Trinity River Basin. Research on the genetic structure of other non-anadromous fish species found in both river systems (marbled sculpin *(Cottus klamathensis)* and Klamath smallscale sucker *(Catostomus rimiculus*)) would be of interest to see if the break at the confluence of the Klamath and Trinity rivers holds for other species.

Based on the level of divergence recovered in both mtDNA and nuclear microsatellites, and the low levels of hybridization detected, taxonomic revision of Klamath speckled dace should be seriously considered. The current taxonomy recognition of all speckled dace in the Klamath-Trinity Basin as (*R o. klamathensis)* may be out of date as it does not reflect the cryptic diversity contained in the Trinity. Trinity speckled dace may in fact represent the first endemic fish species identified from the Trinity River Basin, and should be considered a possible subspecies of speckled dace after further in-depth examination of the lineage. Morphological studies similar to the one performed in (Hoekzema and Sidlauskas 2014) or ecological laboratory studies such as (Harvey et al. 2004) would be useful to further elucidate the Klamath and Trinity lineages' relationship and help guide future taxonomic revision for the species. While Trinity speckled dace are not fully monophyletic due to limited hybridization and possible incomplete lineage sorting with Klamath II, monophyly is not necessarily a requirement for species status under the phylogenetic species concept (Nixon and Wheeler 1990).

Further research is also required on hybridization between the Klamath and Trinity lineages at the confluence of the rivers and in the Salmon River. The contact zone provides an interesting system to study forces responsible for species formation and the maintenance of species in the presence of gene flow. A finer-scaled sampling regime with additional markers specifically designed to study hybridization could be used to evaluate the frequency of different hybrid classes (F1s, F2s, and backcrosses), spatial distribution of hybrids, and the fitness consequences of hybridization. Describing potential

reproductive barriers in speckled dace could also be beneficial to conservation, as the species' high fecundity and colonization ability make it a potentially harmful invasive species. (Kinziger et al 2011; Houston et al. 2012)

These results have important implications for the conservation of freshwater fishes, as native Californian freshwater fishes continue to decline due to human interactions (Moyle et al. 2011). Other systems most likely also harbor cryptic fish diversity, and their identification is important to biodiversity assessments. In the Klamath Trinity basin, four dams are scheduled for removal and ongoing conflicts over water (Jaeger 2004) may strongly affect fish distributions and abundances in the basin. Baseline data will be important for documenting this change and could prove useful for conservation efforts in other drainages (Baumsteiger et al. 2014).

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APPENDIX A

Appendix A: Publically available sequences added to the cyt *b* maximum likelihood tree. The species used (Taxon), the identification code for sequence (ID), the basin where fish was from (Basin), and the paper the sequences were taken from (Source). Kinziger et al. 2011 speckled dace sequences can be found in the Dryad digital depository, the rest of the sequences are from GenBank.

Taxon	Genbank/Dryad ID	Basin/name	Source
R. Osculus	DQ990291	San Gabriel,CA	Dowling et al. 2008
R. Osculus	JN99707	Humboldt, NV	Houston et al. 2012
R. Osculus	DQ990315	Muddy River, NV	Dowling et al. 2008
R. Osculus	DQ990281	N Bonneville , NV	Dowling et al. 2008
R. Osculus	DQ990299	W Bonneville, NV	Dowling et al. 2008
R. Osculus	DQ990287	Gila, AZ	Dowling et al. 2008
R. Osculus	EU158239	Warner, OR	Ardren at al. 2010
R. Osculus	EU158248	Goose Lake, OR	Ardren at al. 2010
R. Osculus	SAC_CAC08	Sacramento, CA	Kinziger et al. 2011
R. Osculus	SAC_CAC10	Sacramento, CA	Kinziger et al. 2011
R. Osculus	SAC_CAC12	Sacramento, CA	Kinziger et al. 2011
R. Osculus	SAC_CLEAR01	Sacramento, CA	Kinziger et al. 2011
R. Osculus	SAC_CLEAR07	Sacramento, CA	Kinziger et al. 2011
R. Osculus	SAC_PIT18	Sacramento, CA	Kinziger et al. 2011
R. Osculus	SAC_PIT15	Sacramento, CA	Kinziger et al. 2011
R. atratulus	AF452078.1	Rogue River, MI	Dowling et al 2002

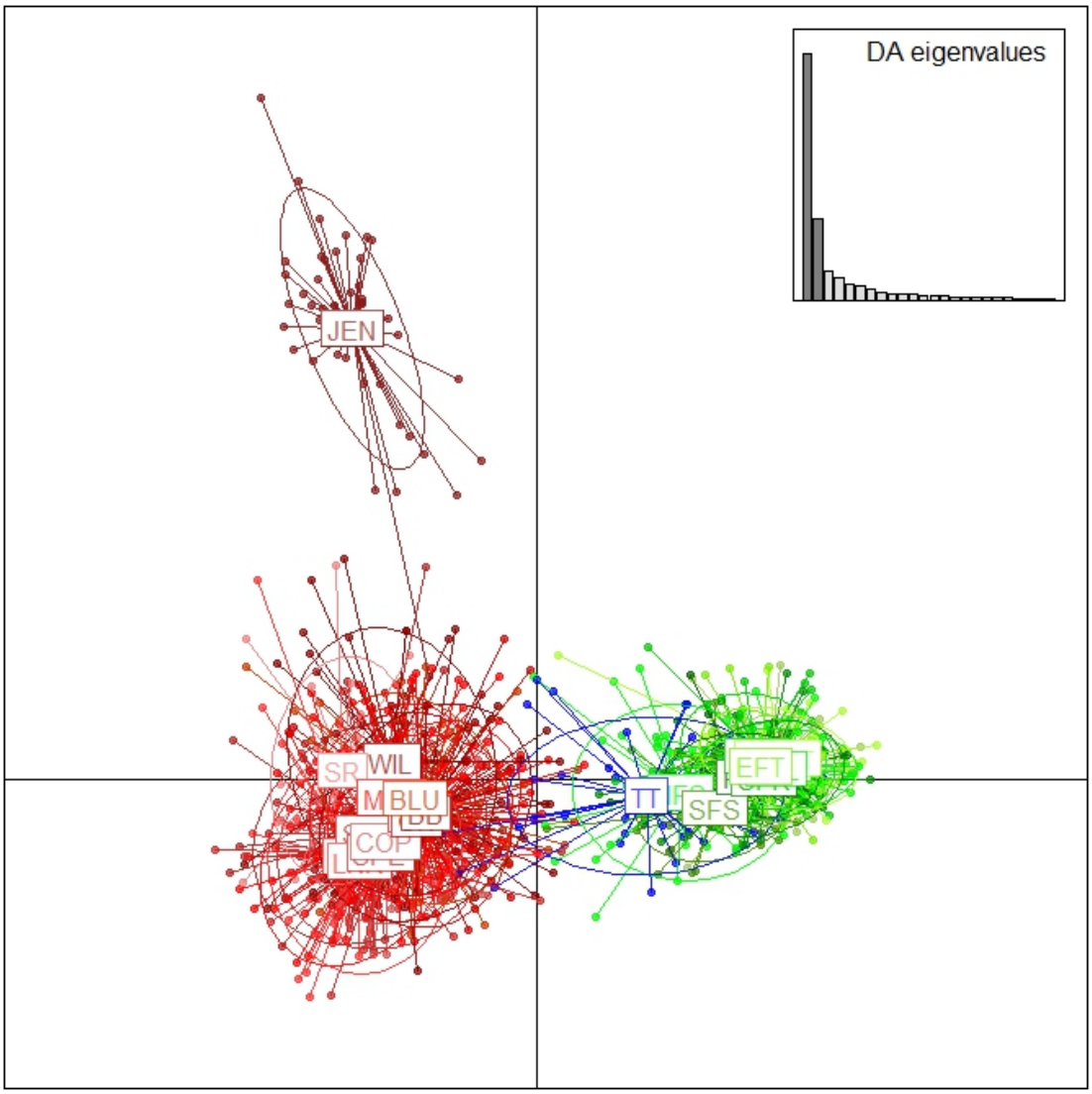
APPENDIX B

Appendix B: Pairwise measures of genetic distance between all 25 populations in the Klamath-Trinity Basin. Microsatellite F_{ST} values above the diagonal calculated in FSTAT, bold values indicate non-significant values. Mitochondrial uncorrected p-distances below the diagonal calculated in MEGA 6.

	SEV	SPR	LNK	SPE	JEN	COP	WIL	SR	SK	MOF	LSR	CN	BB	BLU	NFS	SFS	TT	DL	FG	CAN	GVY	SFK	SWFT	TRL	EFT
SEV	0.000	0.010	0.024	0.016	0.241	0.019	0.026	0.040	0.021	0.032	0.028	0.019	0.039	0.031	0.150	0.182	0.127	0.221	0.204	0.230	0.207	0.222	0.235	0.229	0.190
SPR	0.025	0.000	0.017	0.010	0.245	0.013	0.017	0.031	0.024	0.032	0.020	0.020	0.023	0.022	0.146	0.181	0.122	0.221	0.205	0.233	0.209	0.224	0.234	0.228	0.194
LNK	0.026	0.016	0.000	0.006	0.259	0.014	0.020	0.037	0.024	0.037	0.020	0.016	0.020	0.020	0.147	0.180	0.125	0.211	0.211	0.224	0.202	0.214	0.229	0.220	0.188
SPE	0.027	0.013	0.014	0.000	0.246	**0.002**	**0.005**	0.026	0.011	0.024	**0.007**	**0.002**	0.015	0.011	0.137	0.171	0.114	0.206	0.197	0.220	0.196	0.209	0.223	0.214	0.178
JEN	0.022	0.027	0.021	0.024	0.000	0.248	0.215	0.264	0.218	0.238	0.233	0.227	0.249	0.231	0.285	0.320	0.268	0.329	0.316	0.343	0.312	0.337	0.332	0.345	0.298
COP	0.027	0.010	0.013	0.009	0.026	0.000	**0.004**	0.023	**0.008**	0.016	**0.002**	**0.002**	**0.007**	**0.005**	0.121	0.156	0.103	0.196	0.182	0.207	0.186	0.198	0.209	0.201	0.171
WIL	0.029	0.014	0.015	0.012	0.026	0.008	0.000	0.025	0.017	0.021	**0.008**	**0.009**	0.012	0.011	0.123	0.159	0.106	0.193	0.184	0.208	0.183	0.195	0.209	0.202	0.170
SR	0.032	0.017	0.018	0.014	0.027	0.010	0.008	0.000	0.033	0.021	0.032	0.030	0.037	0.025	0.185	0.224	0.166	0.270	0.253	0.283	0.256	0.275	0.285	0.281	0.238
SK	0.027	0.010	0.013	0.009	0.026	0.006	0.009	0.011	0.000	**0.016**	**0.004**	**0.005**	**0.015**	**0.009**	0.102	0.136	0.087	0.167	0.158	0.178	0.157	0.171	0.179	0.175	0.137
MOF	0.030	0.016	0.016	0.013	0.026	0.009	0.008	0.007	0.010	0.000	**0.012**	0.024	0.031	0.027	0.137	0.173	0.116	0.207	0.190	0.220	0.198	0.213	0.226	0.216	0.179
LSR	0.028	0.013	0.015	0.011	0.026	0.008	0.009	0.010	0.008	0.009	0.000	**0.005**	**0.004**	**0.007**	0.100	0.137	0.085	0.169	0.160	0.180	0.162	0.173	0.184	0.176	0.145
CN	0.027	0.010	0.012	0.009	0.026	0.005	0.008	0.011	0.005	0.010	0.008	0.000	**0.012**	**0.003**	0.106	0.141	0.090	0.177	0.162	0.189	0.167	0.180	0.189	0.183	0.152
BB	0.029	0.013	0.014	0.010	0.028	0.008	0.010	0.011	0.008	0.010	0.010	0.007	0.000	**0.003**	0.099	0.131	0.087	0.176	0.170	0.188	0.167	0.179	0.189	0.183	0.156
BLU	0.029	0.012	0.014	0.010	0.028	0.007	0.010	0.011	0.008	0.010	0.009	0.007	0.008	0.000	0.109	0.143	0.095	0.188	0.172	0.202	0.178	0.192	0.200	0.196	0.165
NFS	0.027	0.009	0.013	0.009	0.026	0.005	0.008	0.010	0.004	0.009	0.007	0.004	0.008	0.007	0.000	**0.010**	0.016	0.054	0.043	0.066	0.052	0.056	0.045	0.048	0.045
SFS	0.027	0.009	0.013	0.009	0.026	0.005	0.007	0.010	0.004	0.008	0.007	0.005	0.008	0.008	0.001	0.000	0.043	0.081	0.075	0.094	0.079	0.080	0.067	0.073	0.069
TT	0.027	0.018	0.018	0.016	0.022	0.014	0.016	0.017	0.015	0.016	0.015	0.014	0.016	0.016	0.014	0.014	0.000	0.056	0.027	0.059	0.052	0.059	0.054	0.047	0.037
DL	0.025	0.028	0.025	0.027	0.014	0.029	0.031	0.032	0.029	0.031	0.029	0.029	0.031	0.030	0.029	0.030	0.020	0.000	0.067	**0.010**	**0.002**	**0.002**	**0.009**	**0.000**	**0.013**
FG	0.025	0.028	0.025	0.027	0.014	0.028	0.031	0.032	0.029	0.031	0.029	0.029	0.030	0.030	0.029	0.030	0.020	0.002	0.000	0.080	0.058	0.069	0.064	0.054	0.054
CAN	0.024	0.028	0.025	0.026	0.013	0.028	0.030	0.032	0.029	0.031	0.029	0.028	0.030	0.029	0.029	0.029	0.020	0.001	0.002	0.000	**0.008**	**0.012**	0.012	**0.006**	**0.019**
GVY	0.025	0.028	0.025	0.027	0.014	0.029	0.031	0.032	0.029	0.032	0.029	0.029	0.031	0.030	0.030	0.030	0.021	0.002	0.003	0.002	0.000	**0.003**	**0.010**	**0.006**	**0.015**
SFK	0.025	0.028	0.025	0.027	0.013	0.029	0.031	0.032	0.029	0.031	0.029	0.029	0.030	0.030	0.029	0.030	0.020	0.001	0.002	0.001	0.001	0.000	**0.008**	**0.000**	**0.018**
SWFT	0.025	0.028	0.025	0.027	0.013	0.029	0.031	0.032	0.029	0.031	0.029	0.029	0.031	0.030	0.029	0.030	0.020	0.001	0.002	0.001	0.001	0.001	0.000	**0.000**	**0.019**
TRL	0.024	0.027	0.025	0.026	0.013	0.028	0.030	0.032	0.029	0.031	0.028	0.028	0.030	0.029	0.029	0.029	0.020	0.001	0.002	0.001	0.001	0.000	0.001	0.000	**0.014**
EFT	0.025	0.028	0.025	0.026	0.013	0.028	0.030	0.032	0.029	0.031	0.029	0.028	0.030	0.029	0.029	0.029	0.020	0.001	0.002	0.001	0.001	0.001	0.001	0.001	**0.000**

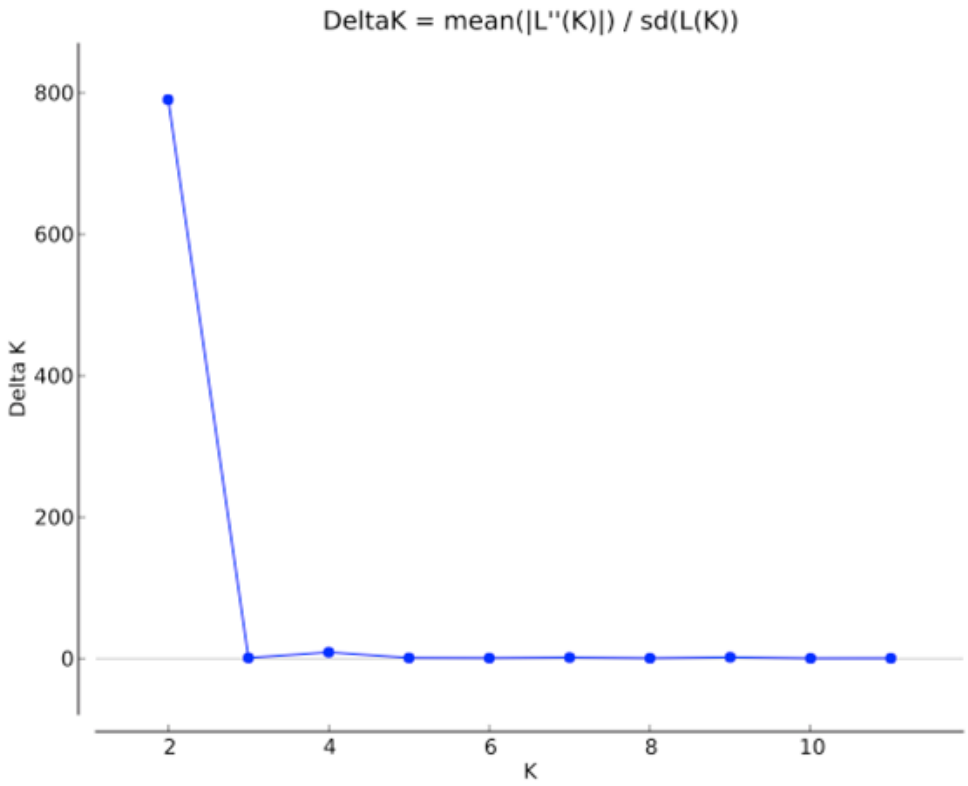
APPENDIX C

Appendix C: DAPC scatter plot with the divergent population Jenny Creek (JEN) included, clusters using population locations as priors. Populations are labeled inside their 95% inertia ellipses and the dots radiating out are individuals (red=Klamath, green=Trinity, Blue=Tish Tang Creek (TT)).



APPENDIX D

Appendix D: Delta K plot resulting from 20 iterations of successive clusters (K =1…12) showing the largest change in K, found at K=2.



APPENDIX E

Appendix E: Haplotype distribution for Klamath-Trinity speckled dace. A total of 78 unique haplotypes were observed in 25 populations.

Haps	BB	BLU	CN	COP	JEN	LNK	LSR	MOF	NFS	SEV	SFS	SK	SPE	SPR	SR	WIL	CAN	DL	FG	GVY	SFK	SWFT	TRL	TT	EFT
Hap1	1					1																		1	
Hap2	1																								
Hap3		2	4	3			1	2	2			1	2			1								2	
Hap4		1		1		1		1							1										
Hap5		1																							
Hap6		1																							
Hap7		2	1	1		1							2		1										
Hap8		1										1			1										
Hap9			1																						
Hap10			1																					1	

Haps	BB	BLU	CN	COP	JEN	LNK	LSR	MOF	NFS	SEV	SFS	SK	SPE	SPR	SR	WIL	CAN	DL	FG	GVY	SFK	SWFT	TRL	TT	EFT
Hap11				1			1																		
Hap12				1																					
Hap13				1																					
Hap14					2																				
Hap15					8																				
Hap16					1																				
Hap17					1	1				2			1												
Hap18						1																			
Hap19						1																			
Hap20						1																			
Hap21							1																		
Hap22							1																		
Hap23							1																		
Hap24							1																		
Hap25							1																		
Hap26								5							7	3									
Hap27									7		9														
Hap28									1																

Haps	BB	BLU	CN	COP	JEN	LNK	LSR	MOF	NFS	SEV	SFS	SK	SPE	SPR	SR	WIL	CAN	DL	FG	GVY	SFK	SWFT	TRL	TT	EFT
Hap29									1																
Hap30										1															
Hap31										2				3											
Hap32										1															
Hap33											1														
Hap34												1													
Hap35												1													
Hap36												1													
Hap37												2													
Hap38												1													
Hap39													1												
Hap40													1												
Hap41													1	1										1	
Hap42													1		1										
Hap43														1											
Hap44														1											
Hap45														1											
Hap46														2											

Haps	BB	BLU	CN	COP	JEN	LNK	LSR	MOF	NFS	SEV	SFS	SK	SPE	SPR	SR	WIL	CAN	DL	FG	GVY	SFK	SWFT	TRL	TT	EFT
Hap47														1											
Hap48															1										
Hap49															1										
Hap50															2										
Hap51															1										
Hap52																1									
Hap53																1								3	
Hap54																1									
Hap55																1									
Hap56																	2								
Hap57																	1					1			
Hap58																	16	4	4	4	7	6	6	4	6
Hap59																	1								
Hap60																	2	1					1		1
Hap61																	1								
Hap62																		1							
Hap63																		1							
Hap64																		1							

Haps	BB	BLU	CN	COP	JEN	LNK	LSR	MOF	NFS	SEV	SFS	SK	SPE	SPR	SR	WIL	CAN	DL	FG	GVY	SFK	SWFT	TRL	TT	EFT
Hap65																									1
Hap66																			1						
Hap67																			3						
Hap68																			4					1	
Hap69																			1						
Hap70																			1						
Hap71																			1						
Hap72																				1					
Hap73																				1					
Hap74																				1					
Hap75																					1				
Hap76																						1			
Hap77																								1	
Hap78																								1	